JavaScript Math Reference

‹ Previous

Next ›

New to JavaScript Math?

SHARE

Visit Our Full JavaScript Math Tutorial

Complete Math Reference

Revised July 2025

Name	Description
abs(x)	Returns the absolute value of x
acos(x)	Returns the arccosine of x, in radians
acosh(x)	Returns the hyperbolic arccosine of x
asin(x)	Returns the arcsine of x, in radians
asinh(x)	Returns the hyperbolic arcsine of x
atan(x)	Returns the arctangent of x as a numeric value between -PI/2 and PI/2 radians
atan2(y, x)	Returns the arctangent of the quotient of its arguments
atanh(x)	Returns the hyperbolic arctangent of x

cbrt(x)	Returns the cubic root of x
ceil(x)	Returns x, rounded upwards to the nearest integer
clz32(x)	Returns the number of leading zeros in a 32-bit binary representation of x
cos(x)	Returns the cosine of x (x is in radians)
cosh(x)	Returns the hyperbolic cosine of x
E	Returns Euler's number (approx. 2.718)
exp(x)	Returns the value of E^x
expm1(x)	Returns the value of E^x minus 1
f16round(x)	Returns x, rounded downwards to the nearest integer
floor(x)	Returns x, rounded downwards to the nearest integer
fround(x)	Returns the nearest (32-bit single precision) float representation of a number
LN2	Returns the natural logarithm of 2 (approx. 0.693)
LN10	Returns the natural logarithm of 10 (approx. 2.302)
log(x)	Returns the natural logarithmof x
log10(x)	Returns the base-10 logarithm of x
LOG10E	Returns the base-10 logarithm of E (approx. 0.434)
log1p(x)	Returns the natural logarithm of 1 + x
log2(x)	Returns the base-2 logarithm of x
LOG2E	Returns the base-2 logarithm of E (approx. 1.442)
max(x1,x2,..)	Returns the number with the highest value
min(x1,x2,..)	Returns the number with the lowest value
PI	Returns PI (approx. 3.14)
pow(x, y)	Returns the value of x to the power of y
random()	Returns a random number between 0 and 1
round(x)	Rounds x to the nearest integer
sign(x)	Returns the sign of a number (checks whether it is positive, negative or zero)
sin(x)	Returns the sine of x (x is in radians)
sinh(x)	Returns the hyperbolic sine of x
sqrt(x)	Returns the square root of x

SQRT1_2	Returns the square root of 1/2 (approx. 0.707)
SQRT2	Returns the square root of 2 (approx. 1.414)
tan(x)	Returns the tangent of an angle
tanh(x)	Returns the hyperbolic tangent of a number
trunc(x)	Returns the integer part of a number (x)

See Also:

Full JavaScript Math Tutorial

Full JavaScript Number Tutorial

Full JavaScript Number Reference

REMOVE ADS

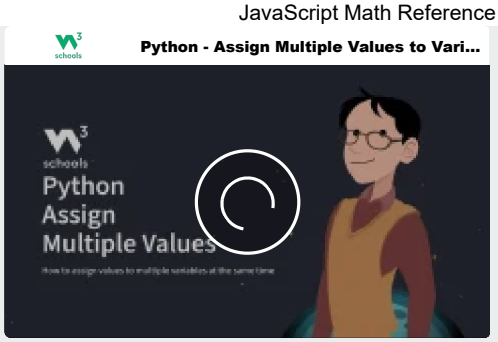
‹ Previous

Sign in to track progress

Next ›



REMOVE ADS



FREE ACCOUNT

Track your progress

XP earned	0
Day streak	0
Spaces	0
League	—

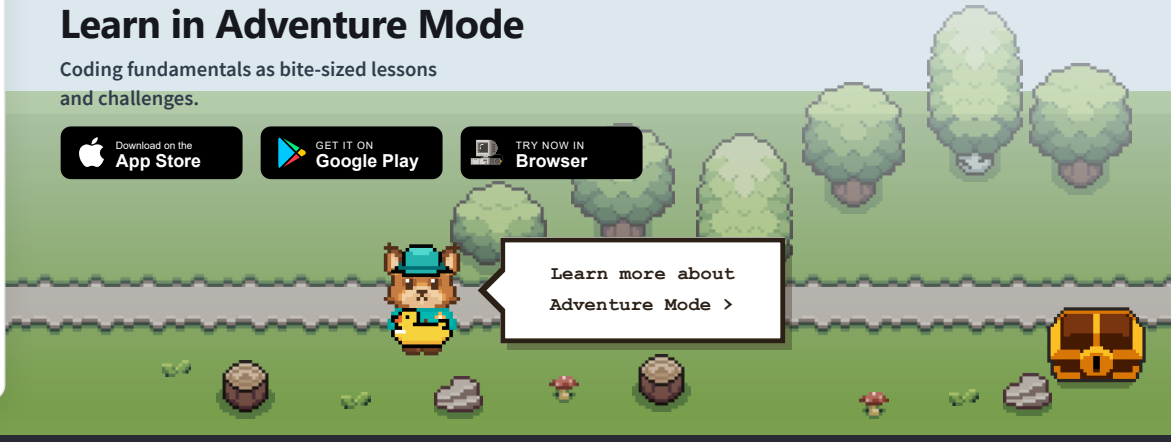
Sign in to track progress

NEW

Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store | GET IT ON Google Play | TRY NOW IN Browser



schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial

PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted

our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Map values()

‹ Previous JavaScript Map Reference Next ›

Map.values()

SHARE

Example

```
// Create a Map
const fruits = new Map([
  ["apples", 500],
  ["bananas", 300],
  ["oranges", 200]
]);

// List all values
let text = "";
for (const x of fruits.values()) {
  text += x;
}
```

Try it Yourself »

More Examples Below !

Description

The `values()` method returns an iterator object with the values in a map.

The `values()` method does not change the original map.

Syntax

```
map.values()
```

Parameters

NONE

Return Value

Type	Description
Iterator	An iterable object with the values of the map.

REMOVE ADS

Related Pages:

JavaScript Maps

JavaScript Iterables

Full JavaScript Map Reference

More Examples

Use the `values()` method to sum the values in a Map:

Example

```
// Create a Map
const fruits = new Map([
  ["apples", 500],
  ["bananas", 300],
  ["oranges", 200]
]);
```

```
// Sum all Values
let total = 0;
for (const x of fruits.values()) {
  total += x;
}
```

Try it Yourself »

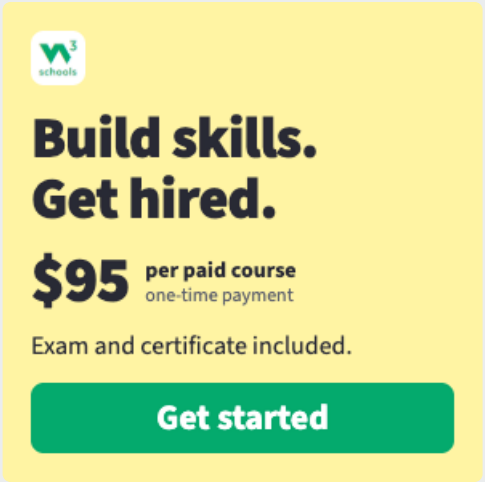
Browser Support

`map.values()` is an ECMAScript6 (ES6 2015) feature.

JavaScript 2015 is supported in all browsers since **June 2017**:

				
Chrome 51	Edge 15	Firefox 54	Safari 10	Opera 38
May 2016	Apr 2017	Jun 2017	Sep 2016	Jun 2016

‹ Previous JavaScript Map Reference Next ›



REMOVE ADS

Python - Introduction - W3Schools.com



FREE ACCOUNT

Track your progress

XP earned 0

Day streak 0

Spaces 0

League —

Sign in to track progress

NEW

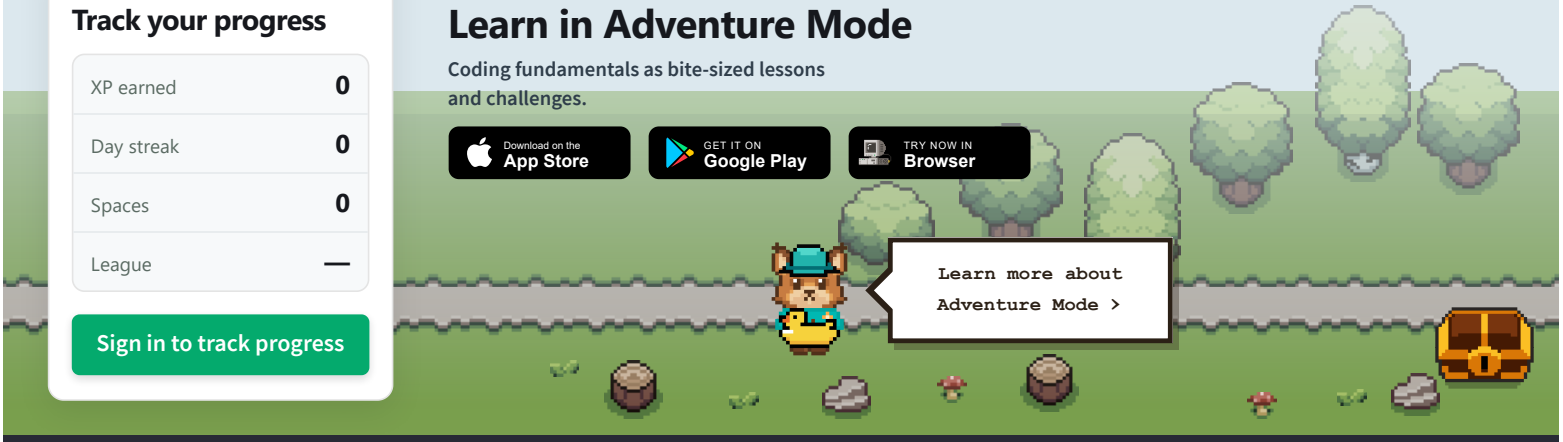
Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store

GET IT ON Google Play

TRY NOW IN Browser



schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial

PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate

FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted

our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Math.abs()

‹ Previous JavaScript Math Object Next ›

Examples

```
let x = Math.abs(-7.25);
```

Try it Yourself »

```
let a = Math.abs(7.25);
let b = Math.abs(-7.25);
let c = Math.abs(null);
let d = Math.abs("Hello");
let e = Math.abs(2-3);
```

Try it Yourself »

Description

SHARE

The `Math.abs()` method returns the absolute value of a number.

JavaScript Rounding Functions

The Math.abs() Method
The Math.ceil() Method

The Math.floor() Method
The Math.round() Method
The Math.fround() Method
The Math.f16round() Method
The Math.trunc() Method

REMOVE ADS

Syntax

```
Math.abs(x)
```

Parameters

Parameter	Description
x	Required. A number.

Return Value

Type	Description
Number	The absolute value of the number. `NaN` if the value is not a number. 0 if the value is `null`.

See Also:

Full JavaScript Math Tutorial

Full JavaScript Math Reference

Full JavaScript Numbers Tutorial

Full JavaScript Number Reference

Browser Support

`Math.abs()` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

Chrome	Edge	Firefox	Safari	Opera

‹ Previous | JavaScript Math Object | Next ›



REMOVE ADS



FREE ACCOUNT

Track your progress

XP earned 0

Day streak 0

Spaces 0

NEW

Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store | GET IT ON Google Play | TRY NOW IN Browser



PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Math.abs()

‹ Previous JavaScript Math Object Next ›

Examples

```
let x = Math.abs(-7.25);
```

Try it Yourself »

```
let a = Math.abs(7.25);
let b = Math.abs(-7.25);
let c = Math.abs(null);
let d = Math.abs("Hello");
let e = Math.abs(2-3);
```

Try it Yourself »

Description

SHARE

The `Math.abs()` method returns the absolute value of a number.

JavaScript Rounding Functions

The Math.abs() Method
The Math.ceil() Method

The Math.floor() Method
The Math.round() Method
The Math.fround() Method
The Math.f16round() Method
The Math.trunc() Method

REMOVE ADS

Syntax

```
Math.abs(x)
```

Parameters

Parameter	Description
x	Required. A number.

Return Value

Type	Description
Number	The absolute value of the number. `NaN` if the value is not a number. 0 if the value is `null`.

See Also:

Full JavaScript Math Tutorial

Full JavaScript Math Reference

Full JavaScript Numbers Tutorial

Full JavaScript Number Reference

Browser Support

`Math.abs()` is an ECMAScript1 (JavaScript 1997) feature.

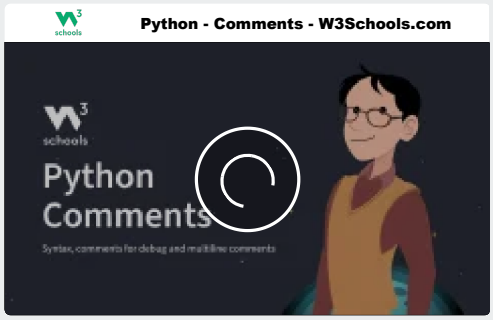
It is supported in all browsers:

Chrome	Edge	Firefox	Safari	Opera

‹ Previous | JavaScript Math Object | Next ›



REMOVE ADS



FREE ACCOUNT
Track your progress
XP earned 0
Day streak 0
Spaces 0

NEW
Learn in Adventure Mode
Coding fundamentals as bite-sized lessons and challenges.
Download on the App Store | GET IT ON Google Play | TRY NOW IN Browser

League —

Sign in to track progress



PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Math.acos()

‹ Previous JavaScript Math Object Next ›

Examples

```
let x = Math.acos(-1);
```

Try it Yourself »

```
let x = Math.acos(0);
```

Try it Yourself »

```
let x = Math.acos(1);
```

Try it Yourself »

Description

SHARE

The `Math.acos()` method returns the arccosine (in radians) of a number.

The `Math.acos()` method returns a value value between 0 and PI.

The `Math.acos()` method expects a parameter in the range -1 to 1.

`Math.acos(-1)` returns PI.

`Math.acos(0)` returns PI/2.

`Math.acos(1)` returns 0.

JavaScript Sine and Cosine Methods:

The Math.sin() Method
The Math.sinh() Method
The Math.asin() Method
The Math.asinh() Method

The Math.cos() Method
The Math.cosh() Method
The Math.acos() Method
The Math.acosh() Method

Syntax

```
Math.acos(x)
```

Parameter	Description
x	Required. A number.

Return Value

Type	Description
Number	0 to PI. `NaN` if the parameter is outside the range of -1 to 1 or not numeric.

See Also:

Full JavaScript Math Tutorial

Full JavaScript Math Reference

Full JavaScript Numbers Tutorial

Full JavaScript Number Reference

REMOVE ADS

Browser Support

`Math.acos()` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

Chrome	Edge	Firefox	Safari	Opera

‹ Previous JavaScript Math Object Next ›



REMOVE ADS



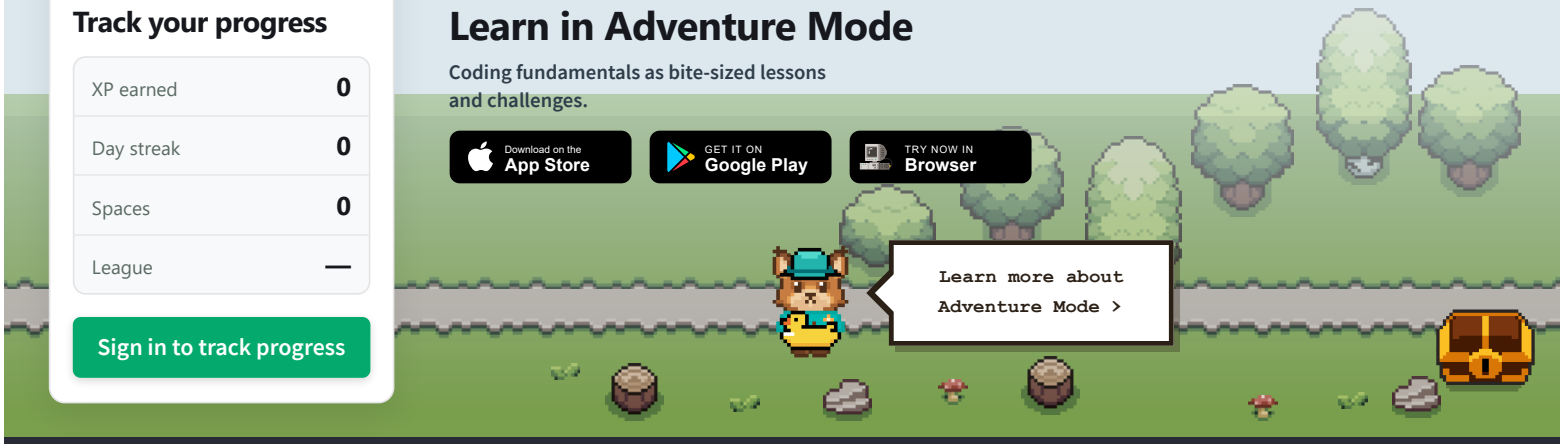
FREE ACCOUNT
Track your progress

XP earned 0
Day streak 0
Spaces 0
League —

Sign in to track progress

NEW
Learn in Adventure Mode
Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store
GET IT ON Google Play
TRY NOW IN Browser



W3 schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference

Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Math.acosh()

‹ Previous JavaScript Math Object Next ›

Examples

```
let x = Math.acosh(1);
```

Try it Yourself »

```
let x = Math.acosh(2);
```

Try it Yourself »

Description

SHARE

The `Math.acosh()` method returns the hyperbolic arc-cosine of a number.

If the parameter is less than 1, the method returns NaN.

JavaScript Sine and Cosine Methods:

The Math.sin() Method
The Math.sinh() Method
The Math.asin() Method
The Math.asinh() Method

The Math.cos() Method
The Math.cosh() Method
The Math.acos() Method
The Math.acosh() Method

Syntax

```
Math.acosh(x)
```

Parameters

Parameter	Description
x	Required. A number.

Return Value

Type	Description
Number	The hyperbolic arc-cosine of the parameter. `NaN` if the parameter is less than 1 or not numeric.

See Also:

Full JavaScript Math Tutorial

Full JavaScript Math Reference

Full JavaScript Numbers Tutorial

Full JavaScript Number Reference

Browser Support

`Math.acosh()` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:







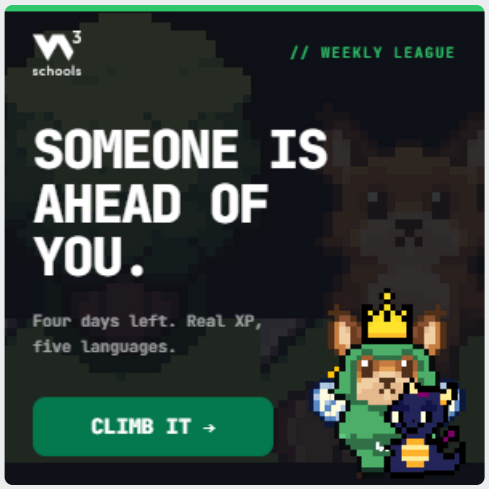
Chrome	Edge	Firefox	Safari	Opera

‹ Previous

JavaScript Math Object

Next ›

REMOVE ADS



REMOVE ADS



FREE ACCOUNT

Track your progress

XP earned 0

Day streak 0

Spaces 0

NEW

Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store | GET IT ON Google Play | TRY NOW IN Browser

League —

Sign in to track progress



PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate

FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Math.asin()

‹ Previous JavaScript Math Object Next ›

Examples

```
let x = Math.asin(-1);
```

Try it Yourself »

```
let x = Math.asin(0);
```

Try it Yourself »

```
let x = Math.asin(1);
```

Try it Yourself »

Description

SHARE

The `Math.asin()` method returns the arcsine (in radians) of a number.

The `Math.asin()` method returns a value between -PI/2 and PI/2.

The `Math.asin()` method expects a parameter in the range -1 to 1.

`Math.asin(-1)` returns -PI/2.

`Math.asin(0)` returns 0.

`Math.asin(1)` returns PI/2.

JavaScript Sine and Cosine Methods:

The Math.sin() Method
The Math.sinh() Method
The Math.asin() Method
The Math.asinh() Method

The Math.cos() Method
The Math.cosh() Method
The Math.acos() Method
The Math.acosh() Method

Syntax

```
Math.asin(x)
```

Parameters

Parameter	Description
x	Required. A number.

Return Value

Type	Description
Number	-PI/2 to PI/2. `NaN` if the parameter is not numeric or outside the range of -1 to 1.

See Also:

Full JavaScript Math Tutorial

Full JavaScript Math Reference

Full JavaScript Numbers Tutorial

Full JavaScript Number Reference

REMOVE ADS

Browser Support

`Math.asin()` is an ECMAScript1 (JavaScript 1997) feature.

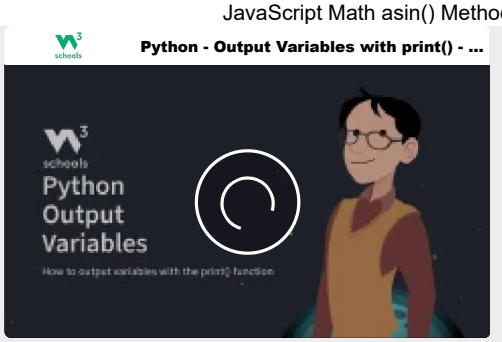
It is supported in all browsers:

Chrome	Edge	Firefox	Safari	Opera

‹ Previous | JavaScript Math Object | Next ›



REMOVE ADS



FREE ACCOUNT

Track your progress

XP earned	0
Day streak	0
Spaces	0
League	—

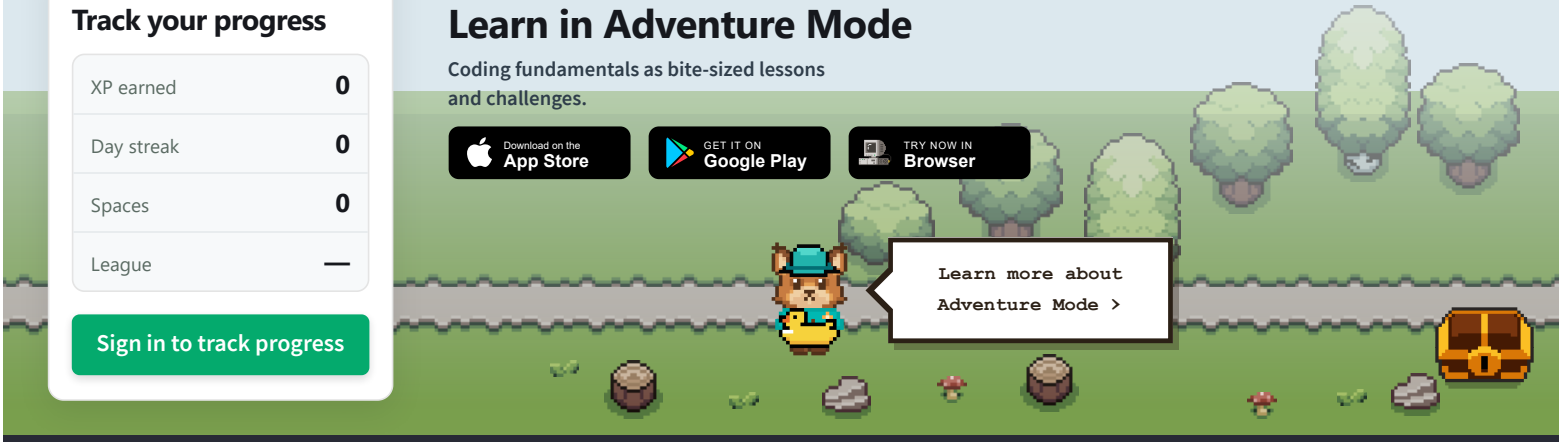
Sign in to track progress

NEW

Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store | GET IT ON Google Play | TRY NOW IN Browser



schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial

PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted

our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Math.asinh()

‹ Previous JavaScript Math Object Next ›

Examples

```
let x = Math.asinh(0);
```

Try it Yourself »

```
let x = Math.asinh(1);
```

Try it Yourself »

Description

SHARE

The `Math.asinh()` method returns the hyperbolic arc-sine of a number.

JavaScript Sine and Cosine Methods:

The Math.sin() Method
The Math.sinh() Method
The Math.asin() Method
The Math.asinh() Method

The Math.cos() Method

The Math.cosh() Method
The Math.acos() Method
The Math.acosh() Method

Syntax

```
Math.asinh(x)
```

Parameters

Parameter	Description
x	Required. A number.

Return Value

Type	Description
Number	The hyperbolic arc-sine of the parameter. `NaN` if the parameter is not numeric.

See Also:

Full JavaScript Math Tutorial

Full JavaScript Math Reference

Full JavaScript Numbers Tutorial

Full JavaScript Number Reference

Browser Support

`Math.asinh()` is an ECMAScript6 (ES6 2015) feature.

JavaScript 2015 is supported in all browsers since **June 2017**:







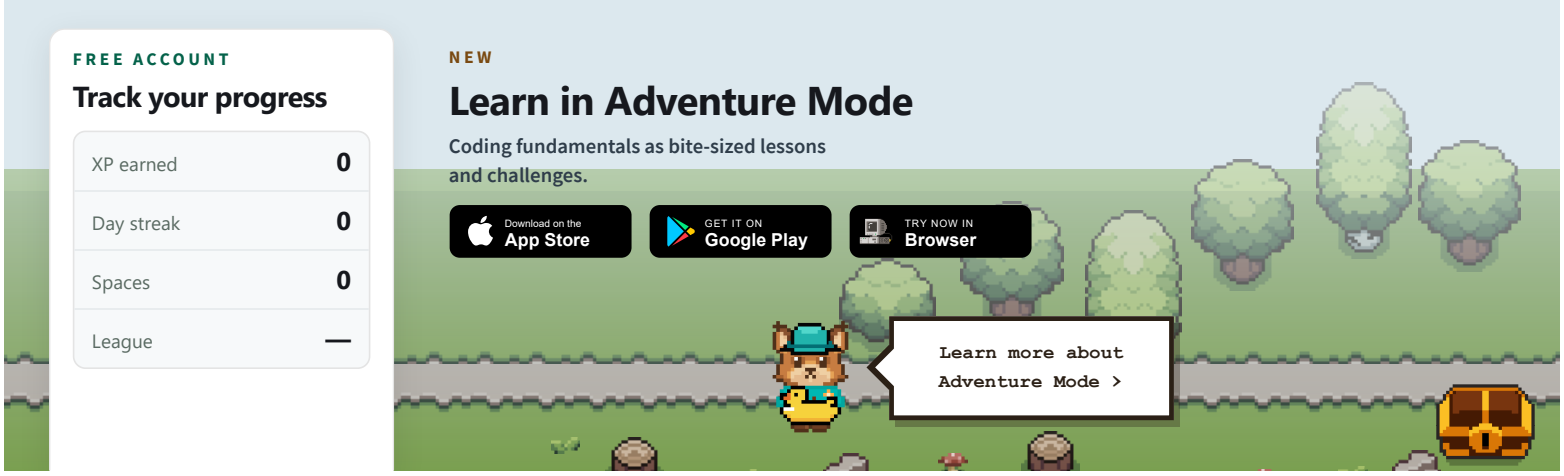
Chrome 51	Edge 15	Firefox 54	Safari 10	Opera 38
May 2016	Apr 2017	Jun 2017	Sep 2016	Jun 2016

‹ Previous JavaScript Math Object Next ›



REMOVE ADS





Sign in to track progress

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples

How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Math.atan()

‹ Previous JavaScript Math Object Next ›

Example

```
let x = Math.atan(2);
```

Try it Yourself »

Description

SHARE

The `Math.atan()` method returns the arctangent of a number as a value between -PI/2 and PI/2 radians.

JavaScript Tangent Methods:

The Math.tan() Method
The Math.tanh() Method
The Math.atan() Method
The Math.atan2() Method
The Math.atanh() Method

Sine and Cosine Methods:

The Math.sin() Method
The Math.cos() Method

Syntax

```
Math.atan(x)
```

Parameters

Parameter	Description
x	Required. A number.

Return Value

Type	Description
Number	-PI to PI/2. `NaN` if the number is empty or not numeric.

See Also:

Full JavaScript Math Tutorial

Full JavaScript Math Reference

Full JavaScript Numbers Tutorial

Full JavaScript Number Reference

Browser Support

`Math.atan()` is an ECMAScript1 (JavaScript 1997) feature.

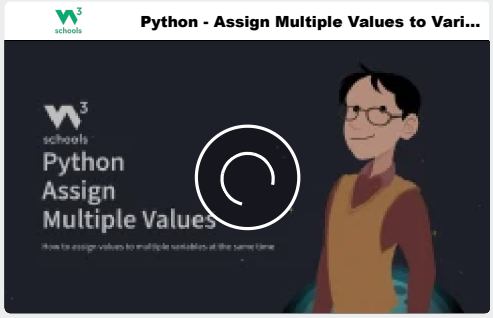
It is supported in all browsers:

Chrome	Edge	Firefox	Safari	Opera

‹ Previous JavaScript Math Object Next ›



REMOVE ADS





schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples

Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Math.atan2()

‹ Previous JavaScript Math Object Next ›

Example - atan2(y,x)

Assume you had a point with the (x,y) coordinates of (4,8), you could calculate the angle in radians between that point and the positive X axis as follows:

```
Math.atan2(8, 4);
```

Try it Yourself »

Description

SHARE

The `Math.atan2()` method returns the arctangent of the quotient of its arguments, as a numeric value between PI and -PI radians.

The number returned represents the counterclockwise angle in radians (not degrees) between the positive X axis and the point (x, y).

Note: With atan2(), the y coordinate is passed as the first argument and the x coordinate is passed as the second argument.

JavaScript Tangent Methods:

The Math.tan() Method
The Math.tanh() Method
The Math.atan() Method
The Math.atan2() Method
The Math.atanh() Method

Sine and Cosine Methods:

The Math.sin() Method
The Math.cos() Method

Syntax

```
Math.atan2(y, x)
```

Parameters

Parameter	Description
y	Required. The y coordinate.
x	Required. The x coordinate.

Return Value

Type	Description
Number	PI to -PI. `NaN` if the values are empty or not numeric.

See Also:

Full JavaScript Math Tutorial

Full JavaScript Math Reference

Full JavaScript Numbers Tutorial

Full JavaScript Number Reference

Browser Support

`Math.atan2()` is an ECMAScript1 (JavaScript 1997) feature.

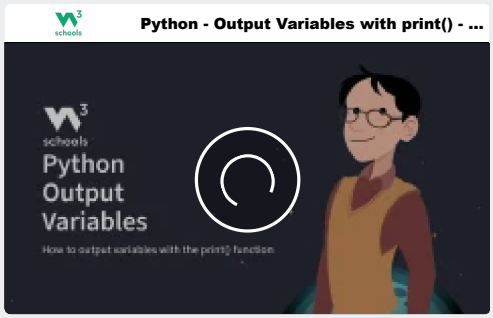
It is supported in all browsers:

Chrome	Edge	Firefox	Safari	Opera

‹ Previous JavaScript Math Object Next ›



REMOVE ADS

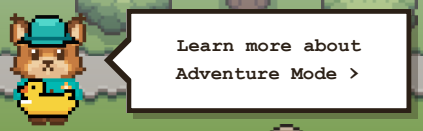


FREE ACCOUNT
Track your progress
XP earned 0

NEW
Learn in Adventure Mode
Coding fundamentals as bite-sized lessons and challenges.

Day streak 0
Spaces 0
League —
Sign in to track progress

Download on the App Store
GET IT ON Google Play
TRY NOW IN Browser



W3schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference

AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Math.atanh()

‹ Previous JavaScript Math Object Next ›

Example

```
let x = Math.atanh(0.5)
```

Try it Yourself »

Description

SHARE

The `Math.atanh()` method returns the hyperbolic arctangent of a number.

The `Math.atanh()` method expects a number between -1 and 1.

The `Math.atanh()` method returns `Infinity` if the number is 1.

The `Math.atanh()` method returns `-Infinity` if the number is -1.

JavaScript Tangent Methods:

The Math.tan() Method
The Math.tanh() Method
The Math.atan() Method
The Math.atan2() Method
The Math.atanh() Method

Sine and Cosine Methods:

The Math.sin() Method
The Math.cos() Method

Syntax

```
Math.atanh(x)
```

Parameters

Parameter	Description
x	Required. A number.

Return Value

Type	Description
Number	The hyperbolic arctangent of the number. Or NaN, or Infinity, or -Infinity. `NaN` if the number is greater than 1, or less than -1.

See Also:

Full JavaScript Math Tutorial

Full JavaScript Math Reference

Full JavaScript Numbers Tutorial

Full JavaScript Number Reference

Browser Support

`Math.atanh()` is an ECMAScript6 (ES6 2015) feature.

JavaScript 2015 is supported in all browsers since **June 2017**:

Chrome	Edge	Firefox	Safari	Opera
51	15	54	10	38
May 2016	Apr 2017	Jun 2017	Sep 2016	Jun 2016

❮ Previous | JavaScript Math Object | Next ❯

Learn it.
Get certified.
Get hired.
$95 per course
one-time payment
Exam and certificate included.
Get started

REMOVE ADS



FREE ACCOUNT
Track your progress
XP earned 0
Day streak 0
Spaces 0
League —

NEW
Learn in Adventure Mode
Coding fundamentals as bite-sized lessons and challenges.
Download on the App Store
GET IT ON Google Play
TRY NOW IN Browser
Learn more about Adventure Mode >

Sign in to track progress



PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples

JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate

FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Math.cbrt()

‹ Previous JavaScript Math Object Next ›

Example

```
let x = Math.cbrt(125);
```

Try it Yourself »

Description

SHARE

The `Math.cbrt()` method returns the cubic root of a number.

See Also:

The Math.sqrt() Method
The Math.SQRT2 Property
The Math.SQRT1_2 Property

Syntax

```
Math.cbrt(x)
```

Parameters

Parameter	Description
x	Required. A number.

Return Value

Type	Description
Number	The cube root of the number.

See Also:

Full JavaScript Math Tutorial

Full JavaScript Math Reference

Full JavaScript Numbers Tutorial

Full JavaScript Number Reference

Browser Support

`Math.cbrt()` is an ECMAScript6 (ES6 2015) feature.

JavaScript 2015 is supported in all browsers since **June 2017**:

Chrome	Edge	Firefox	Safari	Opera
51	15	54	10	38
May 2016	Apr 2017	Jun 2017	Sep 2016	Jun 2016

‹ Previous JavaScript Math Object Next ›

Learn it.
Get certified.
Get hired.

$95 per course
one-time payment

Exam and certificate included.

Get started

REMOVE ADS



FREE ACCOUNT

Track your progress

XP earned 0
Day streak 0
Spaces 0
League —

Sign in to track progress

NEW

Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store
GET IT ON Google Play
TRY NOW IN Browser



schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate

FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Math.ceil()

‹ Previous JavaScript Math Object Next ›

Example

```
Math.ceil(1.4);
```

Try it Yourself »

```
let a = Math.ceil(0.60);
let b = Math.ceil(0.40);
let c = Math.ceil(5);
let d = Math.ceil(5.1);
let e = Math.ceil(-5.1);
let f = Math.ceil(-5.9);
```

Try it Yourself »

Description

SHARE

The `Math.ceil()` method rounds a number rounded UP to the nearest integer.

JavaScript Rounding Functions

The Math.abs() Method
The Math.ceil() Method
The Math.floor() Method
The Math.round() Method
The Math.fround() Method
The Math.f16round() Method
The Math.trunc() Method

REMOVE ADS

Syntax

```
Math.ceil(x)
```

Parameters

Parameter	Description
x	Required. A number.

Return Value

Type	Description
Number	The nearest integer to the number rounding UP.

See Also:

Full JavaScript Math Tutorial

Full JavaScript Math Reference

Full JavaScript Numbers Tutorial

Full JavaScript Number Reference

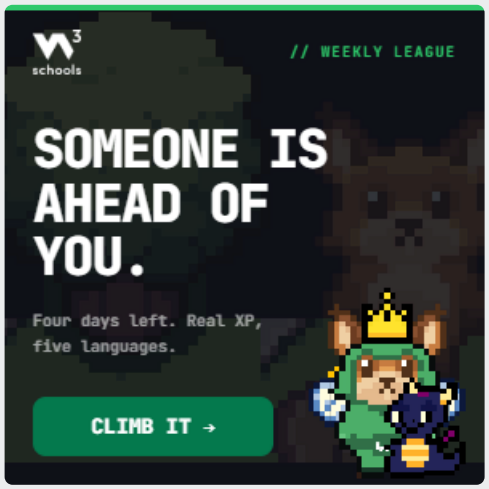
Browser Support

`Math.ceil()` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

Chrome	Edge	Firefox	Safari	Opera

‹ Previous JavaScript Math Object Next ›



REMOVE ADS



FREE ACCOUNT
Track your progress
XP earned 0
Day streak 0
Spaces 0

NEW
Learn in Adventure Mode
Coding fundamentals as bite-sized lessons and challenges.
Download on the App Store
GET IT ON Google Play
TRY NOW IN Browser



PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate

FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Math.clz32()

‹ Previous | JavaScript Math Object | Next ›

Example

```
let a = Math.clz32(0);
let b = Math.clz32(1);
let c = Math.clz32(2);
let d = Math.clz32(4);
```

Try it Yourself »

Description

SHARE

The `Math.clz32()` method returns the number of leading zeros in a 32-bit binary number.

clz32 is short for **CountLeadingZeroes32**.

Syntax

```
Math.clz32(x)
```

Parameters

Parameter	Description

x	Required. A number.

Return Value

Type	Description
Number	Number of leading zeros in the 32-bit binary representation of a number. If the number is 0, it returns 32 (all bits are 0).

See Also:

Full JavaScript Math Tutorial

Full JavaScript Math Reference

Full JavaScript Numbers Tutorial

Full JavaScript Number Reference

Browser Support

`Math.clz32()` is an ECMAScript6 (ES6 2015) feature.

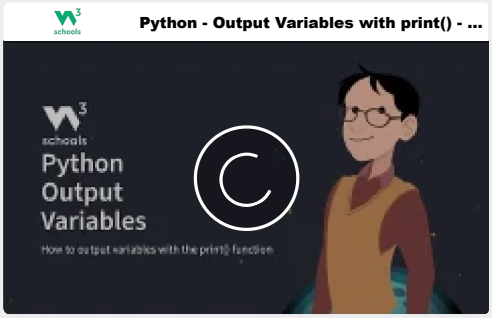
JavaScript 2015 is supported in all browsers since **June 2017**:

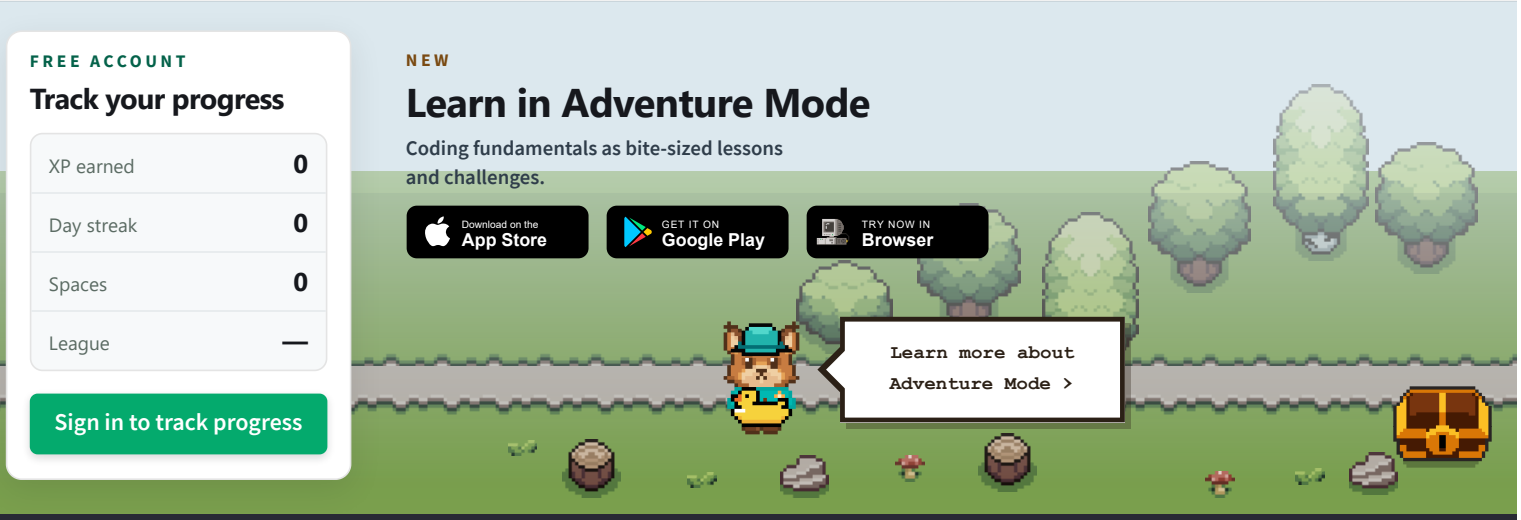
Chrome	Edge	Firefox	Safari	Opera
51	15	54	10	38
May 2016	Apr 2017	Jun 2017	Sep 2016	Jun 2016

‹ Previous JavaScript Math Object Next ›



REMOVE ADS





schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate

FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Math.cos()

‹ Previous JavaScript Math Object Next ›

Examples

```
let x = Math.cos(3.14);
```

Try it Yourself »

```
Math.cos(0);
Math.cos(Math.PI);
Math.cos(2 * Math.PI);
```

Try it Yourself »

Math.cos(x) expects x radians.

To use degrees, convert the degrees to radians.

```
Math.cos(degree * Math.PI / 180);
```

Try it Yourself »

Description

SHARE

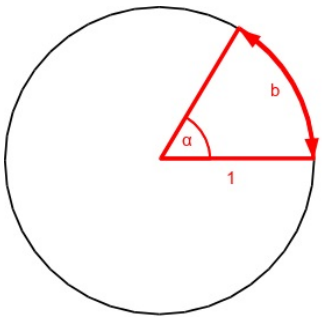
The `Math.cos()` method returns the cosine of a number.

The `Math.cos()` method returns a number between -1 and 1.

The `Math.cos()` method expects the number in radians.

What is Radians?

Radians is an angle's **α** amount of rotation **b** on a circle:



Degrees	PI	Radians
0°	0	0
1°	PI/180	0.0175
30°	PI/6	0.52
45°	PI/4	0.79
90°	PI/2	1.57
180°	PI	3.14
360°	PI*2	6.28

JavaScript Sine and Cosine Methods:

The Math.sin() Method
The Math.sinh() Method
The Math.asin() Method
The Math.asinh() Method

The Math.cos() Method
The Math.cosh() Method
The Math.acos() Method
The Math.acosh() Method

REMOVE ADS

Syntax

```
Math.cos(x)
```

Parameter

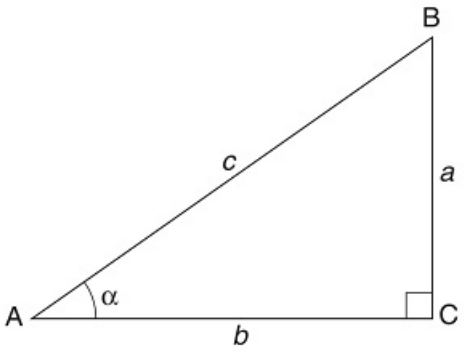
Parameter	Description
x	Required. A number representing radians.

Return Value

Type	Description
Number	-1 to 1 The cosine of the number. `NaN` if the parameter is not numeric.

The Pythagoran Therorem

`Math.sin()`, `Math.cos()`, and `Math.tan()` are related to the Pythagorean theorem:



Theorem: $c^2 = a^2 + b^2$

The sine `Math.sin()` to the angle is **a / c**.

The cosine `Math.cos()` to the angle is **b / c**.

The tangent `Math.tan()` to the angle is **a / b**.

See Also:

Full JavaScript Math Tutorial

Full JavaScript Math Reference

Full JavaScript Numbers Tutorial

Full JavaScript Number Reference

Browser Support

`Math.cos()` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

Chrome	Edge	Firefox	Safari	Opera

‹ Previous | JavaScript Math Object | Next ›



REMOVE ADS

Python - Variable Names - W3Schools.c...



FREE ACCOUNT

Track your progress

XP earned 0

Day streak 0

Spaces 0

League —

Sign in to track progress

NEW

Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store

GET IT ON Google Play

TRY NOW IN Browser



schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial

PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted

our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Math.cosh()

‹ Previous JavaScript Math Object Next ›

Example

```
Math.cosh(3)
```

Try it Yourself »

Description

SHARE

The `Math.cosh()` method returns the hyperbolic cosine of a number.

JavaScript Sine and Cosine Methods:

The Math.sin() Method
The Math.sinh() Method
The Math.asin() Method
The Math.asinh() Method

The Math.cos() Method
The Math.cosh() Method
The Math.acos() Method
The Math.acosh() Method

REMOVE ADS

Syntax

```
Math.cosh(x)
```

Parameters

Parameter	Description
x	Required. A number.

Return Value

Type	Description
Number	The hyperbolic cosine of the number.

See Also:

Full JavaScript Math Tutorial

Full JavaScript Math Reference

Full JavaScript Numbers Tutorial

Full JavaScript Number Reference

Browser Support

`Math.cosh()` is an ECMAScript6 (ES6 2015) feature.

JavaScript 2015 is supported in all browsers since **June 2017**:

				
Chrome 51	Edge 15	Firefox 54	Safari 10	Opera 38
May 2016	Apr 2017	Jun 2017	Sep 2016	Jun 2016

‹ Previous | JavaScript Math Object | Next ›

Learn it.
Get certified.
Get hired.

$95 per course
one-time payment

Exam and certificate included.

Get started

REMOVE ADS



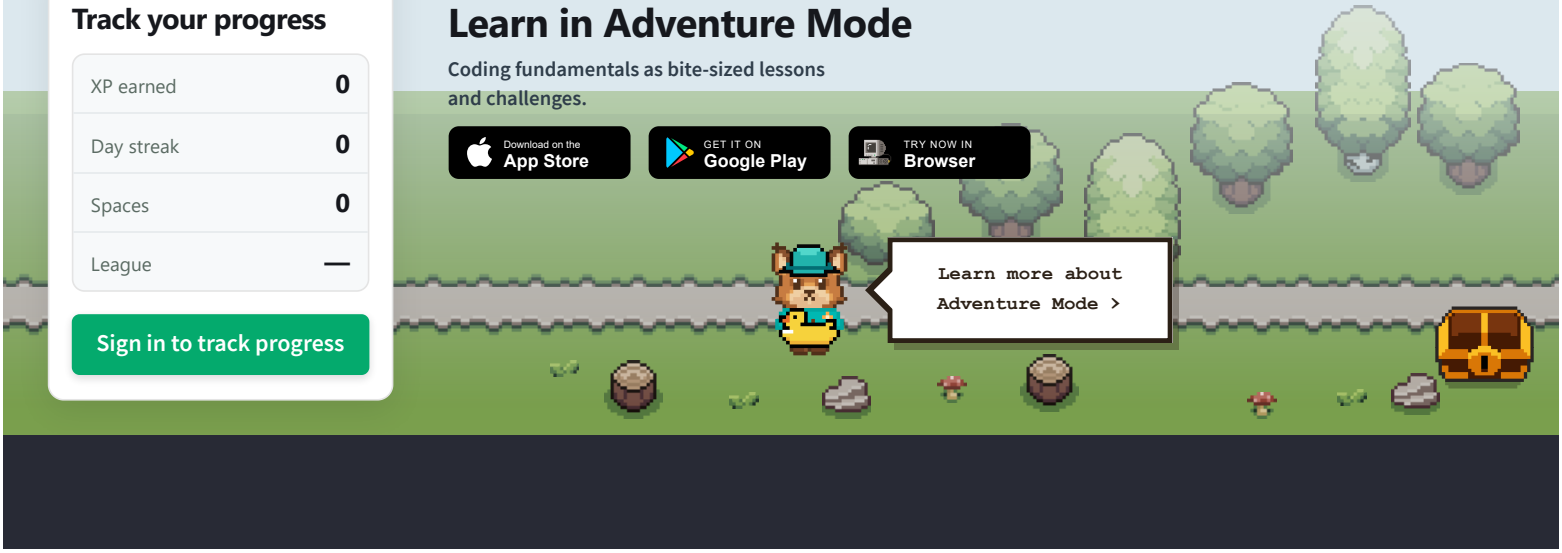
FREE ACCOUNT
Track your progress

XP earned 0
Day streak 0
Spaces 0
League —

Sign in to track progress

NEW
Learn in Adventure Mode
Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store | GET IT ON Google Play | TRY NOW IN Browser



PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples

SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Math.E

‹ Previous JavaScript Math Object Next ›

Example

```
let x = Math.E;
```

Try it Yourself »

Description

SHARE

The `Math.E` property returns Euler's number.

The `Math.E` property returns approximately 2.718.

What is Euler's Number?

Euler's number is the base of natural logarithms.

JavaScript Logarthmic Functions:

The Math.log() Method
The Math.log2() Method

The Math.log10() Method
The Math.log1p() Method

Logarthmic Properies:

The Math.LN2 Property
The Math.LN10 Property
The Math.LOG2E Property
The Math.LOG10E Property

Exponential Functions:

The Math.E Property (Euler's Number)
The Math.exp() Method
The Math.expm1() Method

Syntax

```
Math.E
```

Return Value

Type	Description
Number	Euler's number.

See Also:

Full JavaScript Math Tutorial

Full JavaScript Math Reference

Full JavaScript Numbers Tutorial

Full JavaScript Number Reference

Browser Support

`Math.E` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

Chrome	Edge	Firefox	Safari	Opera

‹ Previous

JavaScript Math Object

Next ›



REMOVE ADS



FREE ACCOUNT

Track your progress

XP earned	0
Day streak	0
Spaces	0
League	—

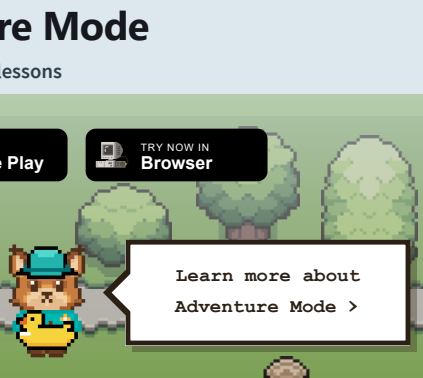
Sign in to track progress

NEW

Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store | GET IT ON Google Play | TRY NOW IN Browser



PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples

How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate

FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Math.exp()

‹ Previous | JavaScript Math Object | Next ›

Examples

```
let x = Math.exp(3);
```

Try it Yourself »

```
let a = Math.exp(1);
let b = Math.exp(-1);
let c = Math.exp(5);
let d = Math.exp(10);
```

Try it Yourself »

Description

SHARE

The `Math.exp()` method returns the value of E^x, where E is Euler's number (approximately 2.7183) and x is the number passed to it.

JavaScript Logarthmic Functions:

The Math.log() Method
The Math.log2() Method

The Math.log10() Method
The Math.log1p() Method

Logarthmic Properies:

The Math.LN2 Property
The Math.LN10 Property
The Math.LOG2E Property
The Math.LOG10E Property

Exponential Functions:

The Math.E Property (Euler's Number)
The Math.exp() Method
The Math.expm1() Method

REMOVE ADS

Syntax

```
Math.exp(x)
```

Parameters

Parameter	Description
x	Required. A number.

Return Value

Type	Description
Number	E^x

See Also:

Full JavaScript Math Tutorial

Full JavaScript Math Reference

Full JavaScript Numbers Tutorial

Full JavaScript Number Reference

Browser Support

`Math.exp()` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

Chrome	Edge	Firefox	Safari	Opera

‹ Previous JavaScript Math Object Next ›

Build skills.
Get hired.
$95 per paid course
one-time payment
Exam and certificate included.
Get started

REMOVE ADS



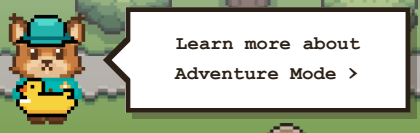
FREE ACCOUNT
Track your progress

XP earned 0
Day streak 0
Spaces 0
League —

Sign in to track progress

NEW
Learn in Adventure Mode
Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store
GET IT ON Google Play
TRY NOW IN Browser



W3 schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference

Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Math.expm1()

‹ Previous JavaScript Math Object Next ›

Examples

```
let x = Math.expm1(3);
```

Try it Yourself »

```
let a = Math.exp(1);
let b = Math.exp(-1);
let c = Math.exp(5);
let d = Math.exp(10);
```

Try it Yourself »

Description

SHARE

The `expm1()` method returns the value of E^x minus 1, where E is Euler's number (approximately 2.7183) and x is the number passed to it.

This method is more accurate than using Math.exp() and subtracting 1.

JavaScript Logarthmic Functions:

The Math.log() Method
The Math.log2() Method
The Math.log10() Method
The Math.log1p() Method

Logarthmic Properies:

The Math.LN2 Property
The Math.LN10 Property
The Math.LOG2E Property
The Math.LOG10E Property

Exponential Functions:

The Math.E Property (Euler's Number)
The Math.exp() Method
The Math.expm1() Method

REMOVE ADS

Syntax

```
Math.expm1(x)
```

Parameters

Parameter	Description
x	Required. A number.

Return Value

Type	Description
Number	E^x minus 1

See Also:

Full JavaScript Math Tutorial

Full JavaScript Math Reference

Full JavaScript Numbers Tutorial

Full JavaScript Number Reference

Browser Support

`Math.expm1()` is an ECMAScript6 (ES6 2015) feature.

JavaScript 2015 is supported in all browsers since **June 2017**:

Chrome	Edge	Firefox	Safari	Opera
51	15	54	10	38
May 2016	Apr 2017	Jun 2017	Sep 2016	Jun 2016

‹ Previous | JavaScript Math Object | Next ›



REMOVE ADS



FREE ACCOUNT

Track your progress

XP earned	0
Day streak	0
Spaces	0
League	—

Sign in to track progress

NEW

Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store | GET IT ON Google Play | TRY NOW IN Browser

Learn more about Adventure Mode >

schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial

PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted

our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Math.f16round()

‹ Previous JavaScript Math Object Next ›

Example

```
let a = Math.f16round(2.60);
let b = Math.f16round(2.50);
let c = Math.f16round(2.49);
let d = Math.f16round(-2.60);
let e = Math.f16round(-2.50);
let f = Math.f16round(-2.49);
```

Try it Yourself »

Description

SHARE

The `Math.f16round()` method returns the nearest 16-bit single precision floating point representation of a number.

JavaScript Rounding Functions

The Math.abs() Method
The Math.ceil() Method
The Math.floor() Method
The Math.round() Method
The Math.fround() Method

The Math.f16round() Method
The Math.trunc() Method

Syntax

```
Math.f16round(x)
```

Parameters

Parameter	Description
x	Required. A number.

Return Value

Type	Description
Number	The nearest 16-bit single precision floating point representation of the number.

See Also:

Full JavaScript Math Tutorial

Full JavaScript Math Reference

Full JavaScript Numbers Tutorial

Full JavaScript Number Reference

Browser Support

`Math.f16round()` is a JavaScript 2025 feature.

ES 2025 is supported in all modern browsers since **May 2025**:

Chrome	Edge	Firefox	Safari	Opera
136	136	129	18.2	120

Apr 2025	Apr 2025	Aug 2024	Des 2024	May 2025

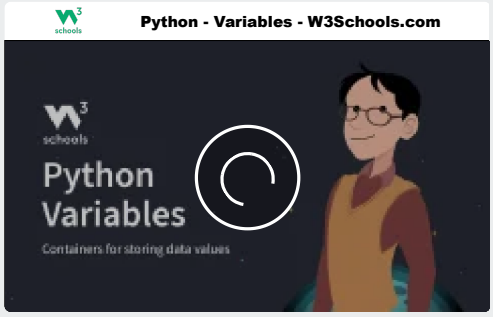
‹ Previous

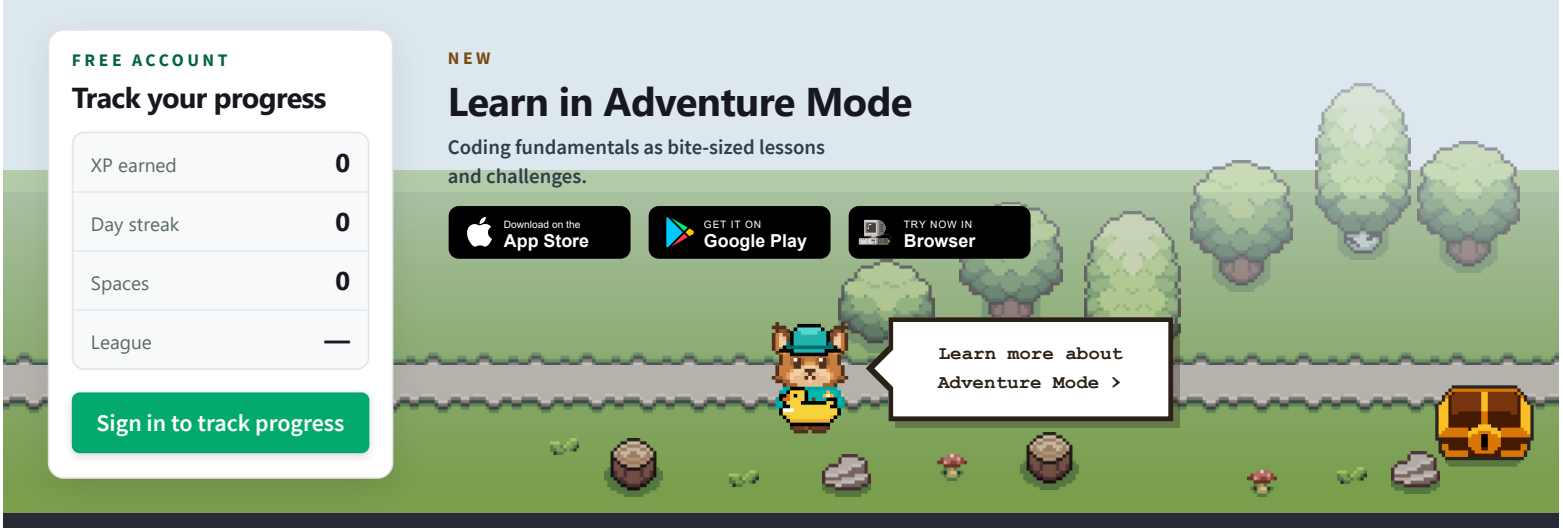
JavaScript Math Object

Next ›



REMOVE ADS





schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples

SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Math.floor()

‹ Previous JavaScript Math Object Next ›

Examples

```
let x = Math.floor(1.6);
```

Try it Yourself »

```
let a = Math.floor(0.60);
let b = Math.floor(0.40);
let c = Math.floor(5);
let d = Math.floor(5.1);
let e = Math.floor(-5.1);
let f = Math.floor(-5.9);
```

Try it Yourself »

Description

SHARE

The `Math.floor()` method rounds a number DOWN to the nearest integer.

JavaScript Rounding Functions

The Math.abs() Method
The Math.ceil() Method
The Math.floor() Method
The Math.round() Method
The Math.fround() Method
The Math.f16round() Method
The Math.trunc() Method

REMOVE ADS

Syntax

```
Math.floor(x)
```

Parameters

Parameter	Description
x	Required. A number.

Return Value

Type	Description
Number	The nearest integer to the number, rounding DOWN.

See Also:

Full JavaScript Math Tutorial

Full JavaScript Math Reference

Full JavaScript Numbers Tutorial

Full JavaScript Number Reference

Browser Support

`Math.floor()` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

Chrome	Edge	Firefox	Safari	Opera

‹ Previous

JavaScript Math Object

Next ›

schools

JS

Get Certified in JavaScript

Boost your career with an official certification

$95

Get certified

REMOVE ADS



FREE ACCOUNT

Track your progress

XP earned 0

Day streak 0

Spaces 0

NEW

Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

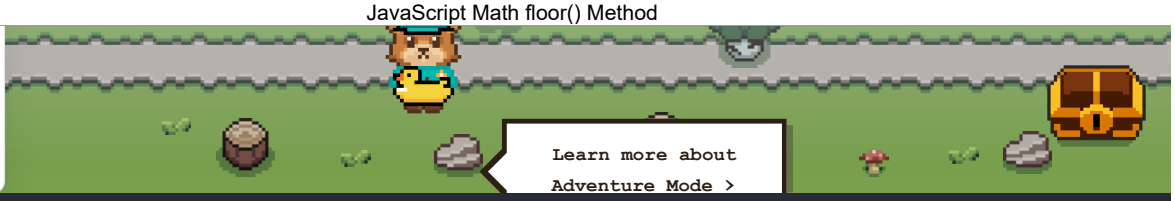
Download on the App Store

GET IT ON Google Play

TRY NOW IN Browser

League —

Sign in to track progress



PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Math.fround()

‹ Previous | JavaScript Math Object | Next ›

Example

```
let a = Math.fround(2.60);
let b = Math.fround(2.50);
let c = Math.fround(2.49);
let d = Math.fround(-2.60);
let e = Math.fround(-2.50);
let f = Math.fround(-2.49);
```

Try it Yourself »

Description

SHARE

The `Math.fround()` method returns the nearest 32-bit single precision float representation of a number.

JavaScript Rounding Functions

The Math.abs() Method
The Math.ceil() Method
The Math.floor() Method
The Math.round() Method
The Math.fround() Method

The Math.f16round() Method
The Math.trunc() Method

Syntax

```
Math.fround(x)
```

Parameters

Parameter	Description
x	Required. A number.

Return Value

Type	Description
Number	The nearest 32-bit single precision float representation of the number.

See Also:

Full JavaScript Math Tutorial

Full JavaScript Math Reference

Full JavaScript Numbers Tutorial

Full JavaScript Number Reference

Browser Support

`Math.fround()` is an ECMAScript6 (ES6 2015) feature.

JavaScript 2015 is supported in all browsers since **June 2017**:

Chrome	Edge	Firefox	Safari	Opera
51	15	54	10	38

May 2016	Apr 2017	Jun 2017	Sep 2016	Jun 2016

‹ Previous

JavaScript Math Object

Next ›



REMOVE ADS



FREE ACCOUNT

Track your progress

XP earned	0
Day streak	0
Spaces	0
League	—

Sign in to track progress

NEW

Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store | GET IT ON Google Play | TRY NOW IN Browser



schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples

SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Math.LN2

‹ Previous JavaScript Math Object Next ›

Example

```
let x = Math.LN2;
```

Try it Yourself »

Description

SHARE

The `Math.LN2` propery returns the natural logarithm of 2.

The `Math.LN2` propery returns approximately 0.693.

JavaScript Logarthmic Functions:

The Math.log() Method
The Math.log2() Method
The Math.log10() Method
The Math.log1p() Method

Logarthmic Properies:

The Math.LN2 Property
The Math.LN10 Property

The Math.LOG2E Property
The Math.LOG10E Property

Exponential Functions:

The Math.E Property (Euler's Number)
The Math.exp() Method
The Math.expm1() Method

Syntax

```
Math.LN2
```

Return Value

Type	Description
Number	The natural logarithm of 2.

See Also:

Full JavaScript Math Tutorial

Full JavaScript Math Reference

Full JavaScript Numbers Tutorial

Full JavaScript Number Reference

Browser Support

`Math.LN2` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

Chrome	Edge	Firefox	Safari	Opera

‹ Previous JavaScript Math Object Next ›

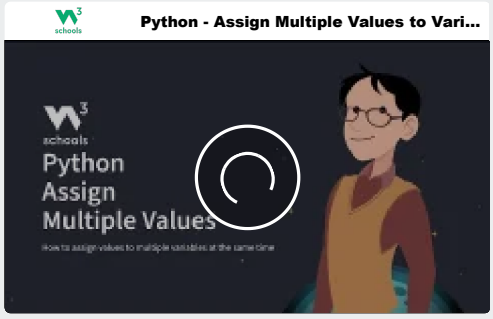
Build skills.
Get hired.

$95 per paid course
one-time payment

Exam and certificate included.

Get started

REMOVE ADS



FREE ACCOUNT

Track your progress

XP earned 0
Day streak 0
Spaces 0
League —

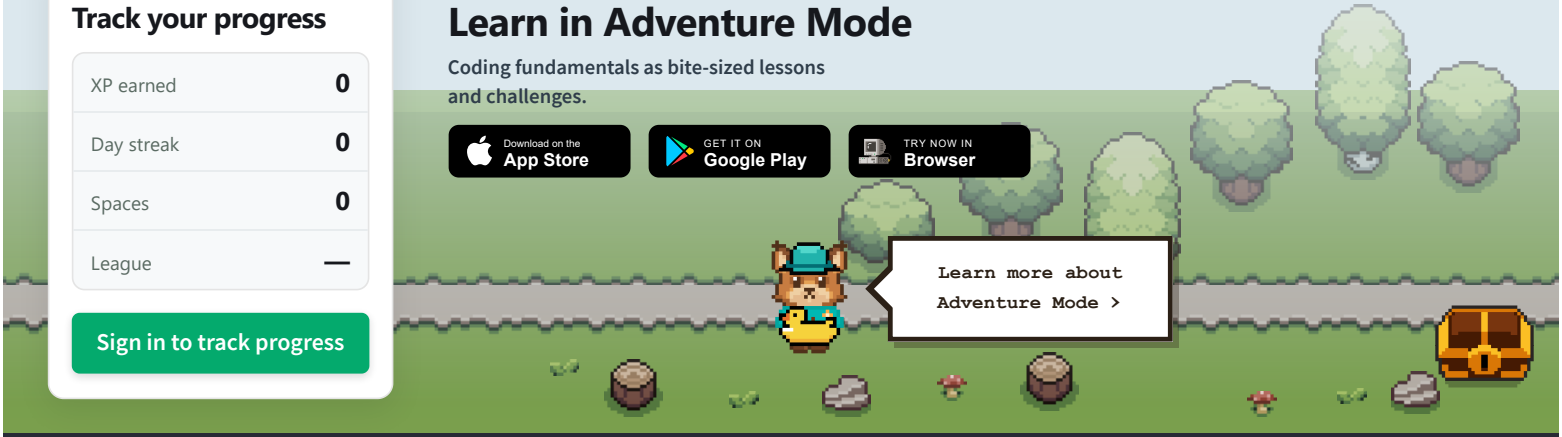
Sign in to track progress

NEW

Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store
GET IT ON Google Play
TRY NOW IN Browser



schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples

Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Math.LN10

‹ Previous JavaScript Math Object Next ›

Example

```
let x = Math.LN10;
```

Try it Yourself »

Description

SHARE

The `Math.LN10` property returns the natural logarithm of 10.

The `Math.LN10` property approximately 2.302.

JavaScript Logarthmic Functions:

The Math.log() Method
The Math.log2() Method
The Math.log10() Method
The Math.log1p() Method

Logarthmic Properies:

The Math.LN2 Property
The Math.LN10 Property

The Math.LOG2E Property
The Math.LOG10E Property

Exponential Functions:

The Math.E Property (Euler's Number)
The Math.exp() Method
The Math.expm1() Method

Syntax

```
Math.LN10
```

Return Value

Type	Description
Number	The natural logarithm of 10.

See Also:

Full JavaScript Math Tutorial

Full JavaScript Math Reference

Full JavaScript Numbers Tutorial

Full JavaScript Number Reference

Browser Support

`Math.LN10` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

Chrome	Edge	Firefox	Safari	Opera

‹ Previous JavaScript Math Object Next ›



REMOVE ADS



FREE ACCOUNT

Track your progress

XP earned	0
Day streak	0
Spaces	0
League	—

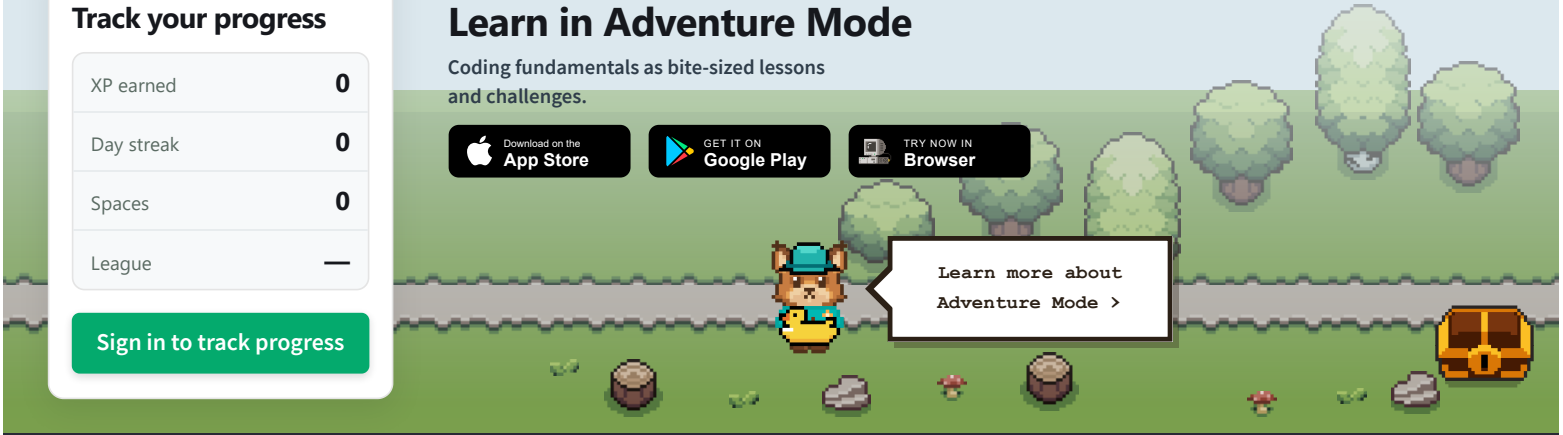
Sign in to track progress

NEW

Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store | GET IT ON Google Play | TRY NOW IN Browser



schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples

Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate

FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Math.log()

‹ Previous JavaScript Math Object Next ›

Examples

```
let x = Math.log(2);
```

Try it Yourself »

```
let a = Math.log(-0);
let b = Math.log(0);
let c = Math.log(-1);
let d = Math.log(1);
let e = Math.log(10);
let f = Math.log(Infinity);
let g = Math.log(-Infinity);
```

Try it Yourself »

Description

SHARE

The `Math.log()` method returns the natural logarithm (base E) of a number.

JavaScript Logarthmic Functions:

The Math.log() Method
The Math.log2() Method
The Math.log10() Method
The Math.log1p() Method

Logarthmic Properies:

The Math.LN2 Property
The Math.LN10 Property
The Math.LOG2E Property
The Math.LOG10E Property

Exponential Functions:

The Math.E Property (Euler's Number)
The Math.exp() Method
The Math.expm1() Method

REMOVE ADS

Syntax

```
Math.log(x)
```

Parameters

Parameter	Description
x	Required. A number.

Return Value

Type	Description
Number	The natural logarithm of the number. `NaN` if the number is negative. `-Infinity` if the number is 0.

See Also:

Full JavaScript Math Tutorial

Full JavaScript Math Reference

Full JavaScript Numbers Tutorial

Full JavaScript Number Reference

Browser Support

`Math.log()` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

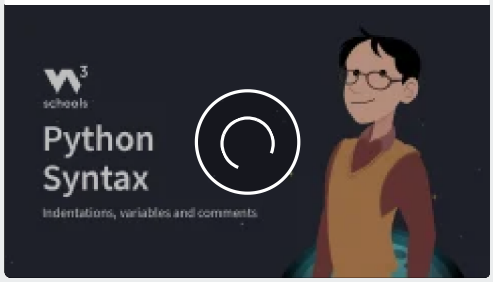
Chrome	Edge	Firefox	Safari	Opera

‹ Previous JavaScript Math Object Next ›



REMOVE ADS

Python - Syntax - W3Schools.com



FREE ACCOUNT

Track your progress

XP earned 0

Day streak 0

Spaces 0

League —

Sign in to track progress

NEW

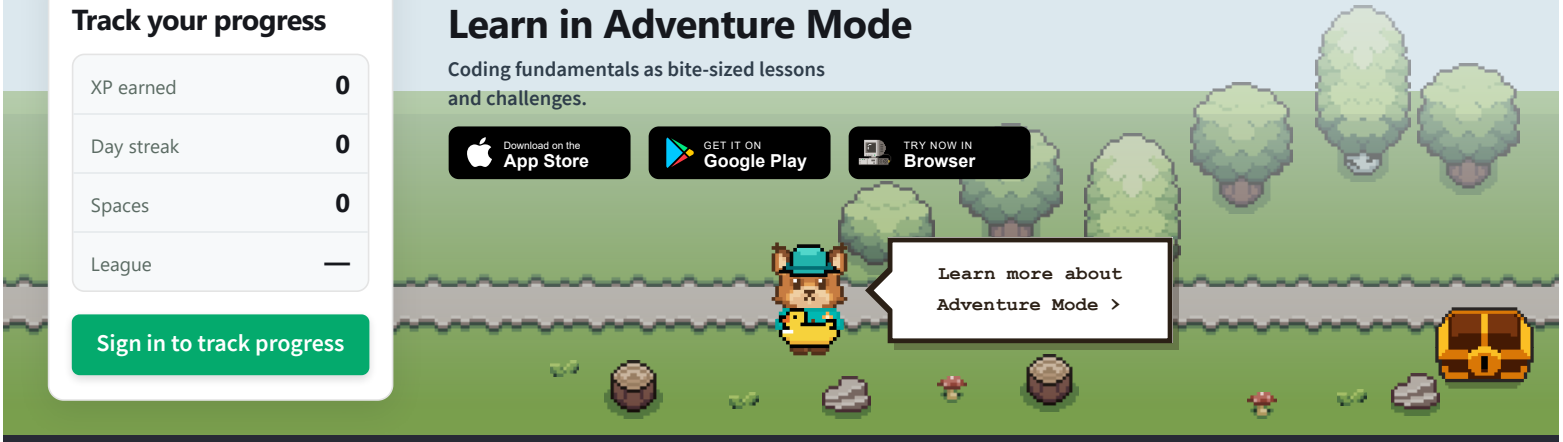
Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store

GET IT ON Google Play

TRY NOW IN Browser



schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial

PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted

our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Math.log10()

‹ Previous JavaScript Math Object Next ›

Examples

```
let x = Math.log10(2);
```

Try it Yourself »

```
let a = Math.log10(2.7183);
let b = Math.log10(2);
let c = Math.log10(1);
let d = Math.log10(0);
let e = Math.log10(-1);
```

Try it Yourself »

Description

SHARE

The `Math.log10()` method returns the base-10 logarithm of a number.

JavaScript Logarthmic Functions:

The Math.log() Method
The Math.log2() Method

The Math.log10() Method
The Math.log1p() Method

Logarthmic Properies:

The Math.LN2 Property
The Math.LN10 Property
The Math.LOG2E Property
The Math.LOG10E Property

Exponential Functions:

The Math.E Property (Euler's Number)
The Math.exp() Method
The Math.expm1() Method

REMOVE ADS

Syntax

```
Math.log10(x)
```

Parameters

Parameter	Description
x	Required. A number.

Return Value

Type	Description
Number	The base-10 logarithm of the number. `NaN` if the number is negative. `-Infinity` if the number is 0.

See Also:

Full JavaScript Math Tutorial

Full JavaScript Math Reference

Full JavaScript Numbers Tutorial

Full JavaScript Number Reference

Browser Support

`Math.log10()` is an ECMAScript6 (ES6 2015) feature.

JavaScript 2015 is supported in all browsers since **June 2017**:

				
Chrome 51	Edge 15	Firefox 54	Safari 10	Opera 38
May 2016	Apr 2017	Jun 2017	Sep 2016	Jun 2016

‹ Previous

JavaScript Math Object

Next ›

Learn it.
Get certified.
Get hired.

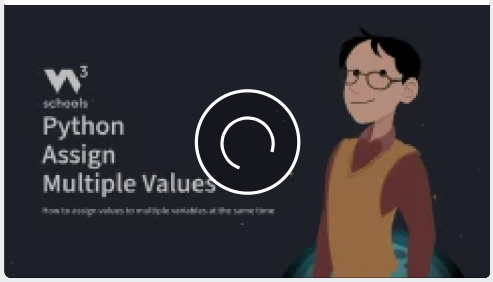
$95 per course
one-time payment

Exam and certificate included.

Get started

REMOVE ADS

Python - Assign Multiple Values to Vari...



FREE ACCOUNT

Track your progress

XP earned 0

Day streak 0

Spaces 0

League —

Sign in to track progress

NEW

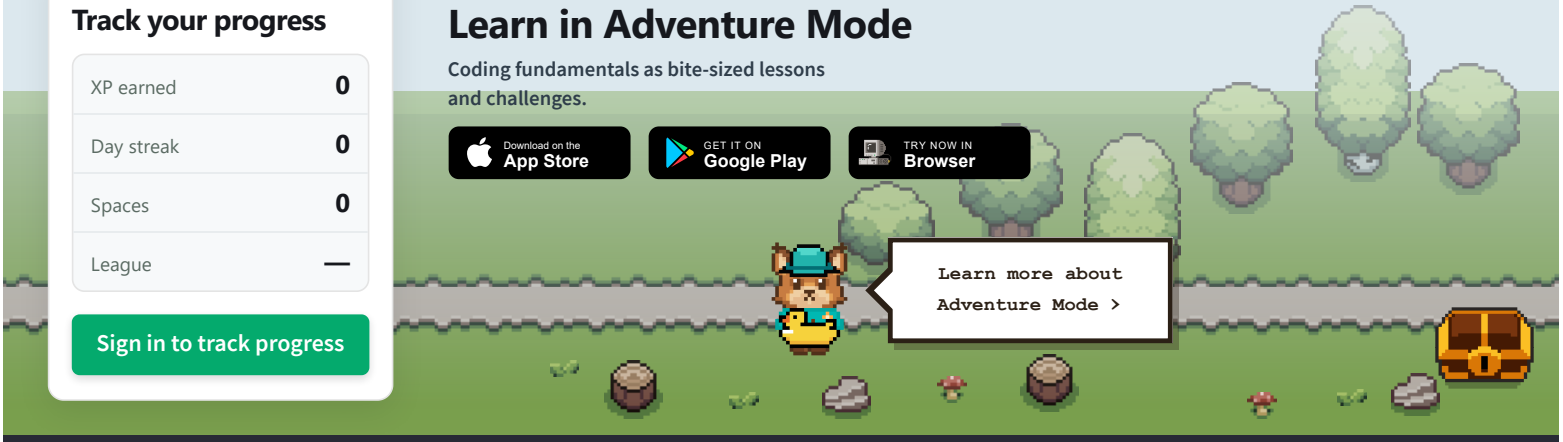
Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store

GET IT ON Google Play

TRY NOW IN Browser



schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial

PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted

our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Math.LOG10E

‹ Previous JavaScript Math Object Next ›

Example

```
let x = Math.LOG10E;
```

Try it Yourself »

Description

SHARE

The `Math.LOG10E` property returns the base-10 logarithm of E.

The `Math.LOG10E` property returns approximately 0.434.

JavaScript Logarthmic Functions:

The Math.log() Method
The Math.log2() Method
The Math.log10() Method
The Math.log1p() Method

Logarthmic Properies:

The Math.LN2 Property
The Math.LN10 Property

The Math.LOG2E Property
The Math.LOG10E Property

Exponential Functions:

The Math.E Property (Euler's Number)
The Math.exp() Method
The Math.expm1() Method

Syntax

```
Math.LOG10E
```

Return Value

Type	Description
Number	The base-10 logarithm of E.

See Also:

Full JavaScript Math Tutorial

Full JavaScript Math Reference

Full JavaScript Numbers Tutorial

Full JavaScript Number Reference

Browser Support

`Math.LOG10E` is an ECMAScript1 (JavaScript 1997) feature.

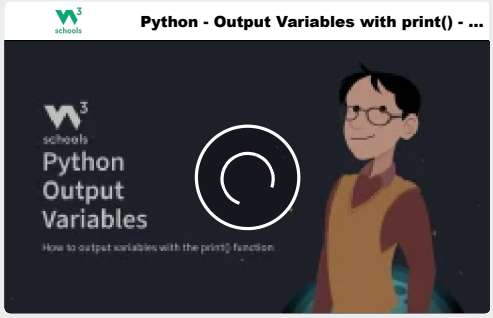
It is supported in all browsers:

Chrome	Edge	Firefox	Safari	Opera

‹ Previous JavaScript Math Object Next ›



REMOVE ADS



FREE ACCOUNT

Track your progress

XP earned	0
Day streak	0
Spaces	0
League	—

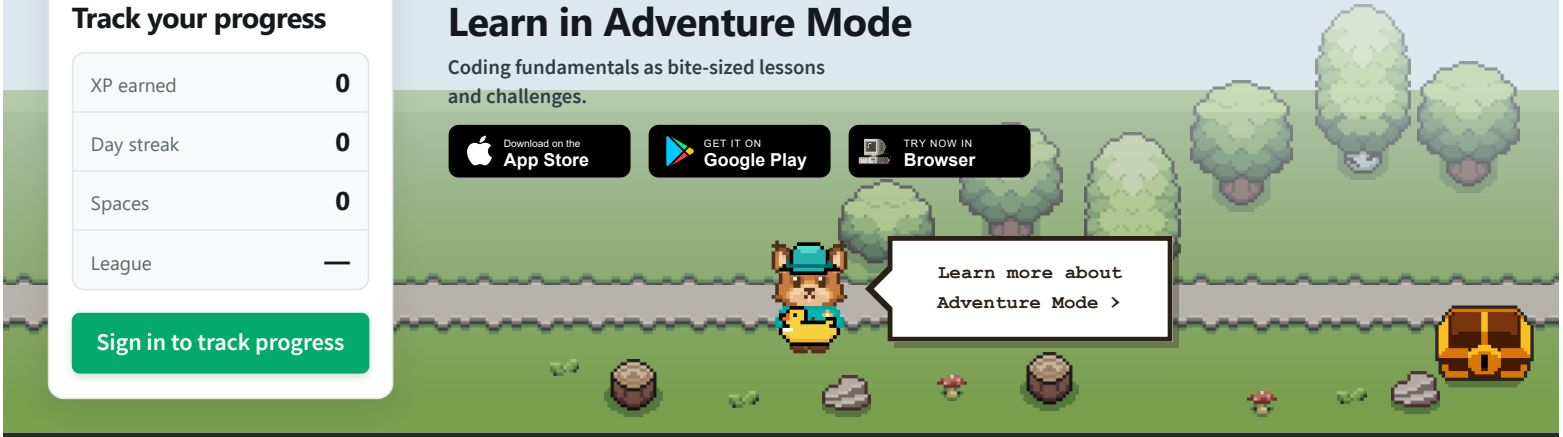
Sign in to track progress

NEW

Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store | GET IT ON Google Play | TRY NOW IN Browser



schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples

Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate

FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Math.log1p()

‹ Previous | JavaScript Math Object | Next ›

Examples

```
let a = Math.log1p(2.7183);
let b = Math.log1p(2);
let c = Math.log1p(1);
let d = Math.log1p(0);
let e = Math.log1p(-1);
```

Try it Yourself »

Description

SHARE

The `Math.log1p()` method returns the natural logarithm (base E) of 1 + a number.

JavaScript Logarthmic Functions:

The Math.log() Method
The Math.log2() Method
The Math.log10() Method
The Math.log1p() Method

Logarthmic Properies:

The Math.LN2 Property
The Math.LN10 Property
The Math.LOG2E Property
The Math.LOG10E Property

Exponential Functions:

The Math.E Property (Euler's Number)
The Math.exp() Method
The Math.expm1() Method

Syntax

```
Math.log1p(x)
```

Parameters

Parameter	Description
x	Required. A number.

Return Value

Type	Description
Number	The natural logarithm (base E) of 1 + the number. `NaN` if the number is less than -1. `-Infinity` if the number is -1.

See Also:

Full JavaScript Math Tutorial

Full JavaScript Math Reference

Full JavaScript Numbers Tutorial

Full JavaScript Number Reference

Browser Support

`Math.log1p()` is an ECMAScript6 (ES6 2015) feature.

JavaScript 2015 is supported in all browsers since **June 2017**:

Chrome	Edge	Firefox	Safari	Opera
51	15	54	10	38
May 2016	Apr 2017	Jun 2017	Sep 2016	Jun 2016

‹ Previous | JavaScript Math Object | Next ›

Build skills.
Get hired.
$95 per paid course
one-time payment
Exam and certificate included.
Get started

REMOVE ADS

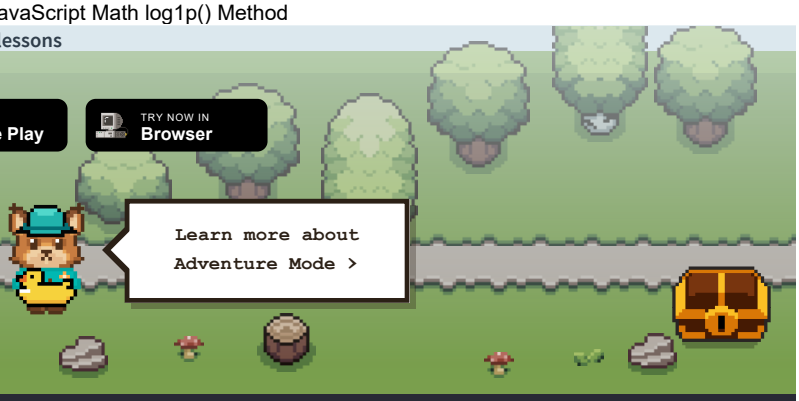


FREE ACCOUNT
Track your progress

NEW
Learn in Adventure Mode

XP earned 0
Day streak 0
Spaces 0
League —
Sign in to track progress

Coding fundamentals as bite-sized lessons and challenges.
Download on the App Store
GET IT ON Google Play
TRY NOW IN Browser



W3schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors

Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate

FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Math.log2()

‹ Previous JavaScript Math Object Next ›

Examples

```
let x = Math.log2(3);
```

Try it Yourself »

```
let a = Math.log2(-Infinity);
let b = Math.log2(Infinity);
let c = Math.log2(-0);
let d = Math.log2(0);
let e = Math.log2(-1);
let f = Math.log2(1);
let g = Math.log2(2);
```

Try it Yourself »

```
let a = Math.log2(2);
let b = Math.log2(4);
let c = Math.log2(8);
let d = Math.log2(16);
let e = Math.log2(32);
let f = Math.log2(64);
```

Try it Yourself »

Description

SHARE

The `Mat.log2()` method returns the base-2 logarithm of a number.

JavaScript Logarthmic Functions:

The Math.log() Method
The Math.log2() Method
The Math.log10() Method
The Math.log1p() Method

Logarthmic Properies:

The Math.LN2 Property
The Math.LN10 Property
The Math.LOG2E Property
The Math.LOG10E Property

Exponential Functions:

The Math.E Property (Euler's Number)
The Math.exp() Method
The Math.expm1() Method

Syntax

```
Math.log2(x)
```

Parameters

Parameter	Description
x	Required. A number.

Return Value

Type	Description

Number	The base-2 logarithm of the number. `NaN` if the number is less than -1. `-Infinity` if the number is -1.

See Also:

Full JavaScript Math Tutorial

Full JavaScript Math Reference

Full JavaScript Numbers Tutorial

Full JavaScript Number Reference

Browser Support

`Math.log2()` is an ECMAScript6 (ES6 2015) feature.

JavaScript 2015 is supported in all browsers since **June 2017**:

				
Chrome 51	Edge 15	Firefox 54	Safari 10	Opera 38
May 2016	Apr 2017	Jun 2017	Sep 2016	Jun 2016

‹ Previous JavaScript Math Object Next ›

Get Certified in JavaScript

Boost your career with an official certification

$95

Get certified

REMOVE ADS



FREE ACCOUNT

Track your progress

XP earned 0

Day streak 0

Spaces 0

League —

Sign in to track progress

NEW

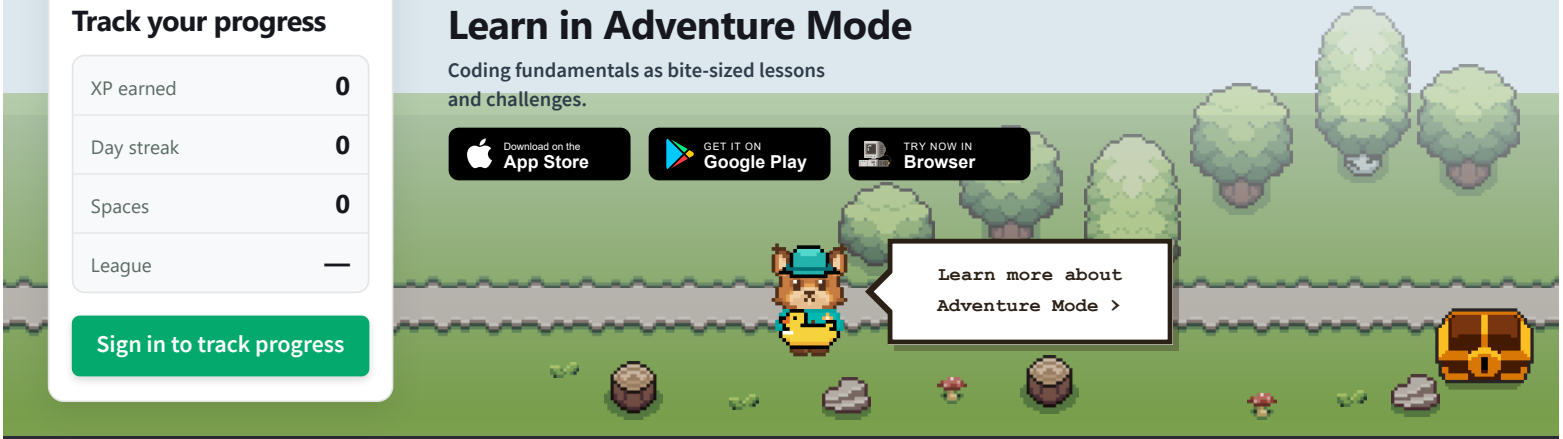
Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store

GET IT ON Google Play

TRY NOW IN Browser



schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate

FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Math LOG2E

‹ Previous JavaScript Math Object Next ›

Example

```
let x = Math.LOG2E;
```

Try it Yourself »

Description

SHARE

The `Math.LOG2E` property returns the base-2 logarithm of E.

The `Math.LOG2E` property returns approximately 1.442.

JavaScript Logarthmic Functions:

The Math.log() Method
The Math.log2() Method
The Math.log10() Method
The Math.log1p() Method

Logarthmic Properies:

The Math.LN2 Property
The Math.LN10 Property

The Math.LOG2E Property
The Math.LOG10E Property

Exponential Functions:

The Math.E Property (Euler's Number)
The Math.exp() Method
The Math.expm1() Method

Syntax

```
Math.LOG2E
```

Return Value

Type	Description
Number	The base-2 logarithm of E.

See Also:

Full JavaScript Math Tutorial

Full JavaScript Math Reference

Full JavaScript Numbers Tutorial

Full JavaScript Number Reference

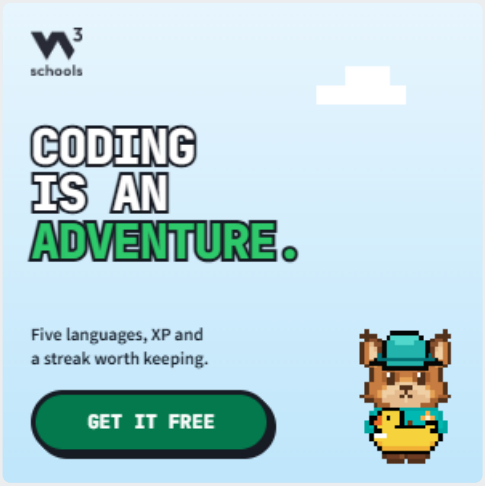
Browser Support

`Math.LOG2E` is an ECMAScript1 (JavaScript 1997) feature.

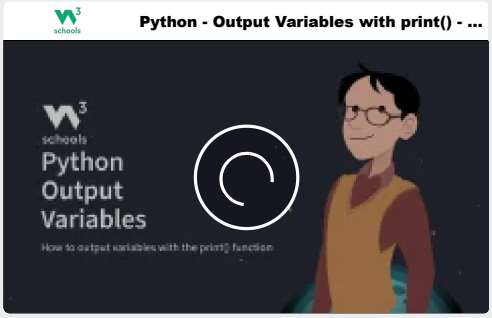
It is supported in all browsers:

Chrome	Edge	Firefox	Safari	Opera

‹ Previous JavaScript Math Object Next ›



REMOVE ADS





schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples

Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Math.max()

‹ Previous JavaScript Math Object Next ›

Example

```
let x = Math.max(5, 10);
```

Try it Yourself »

More Examples

```
let a = Math.max(5, 10);
let b = Math.max(0, 150, 30, 20, 38);
let c = Math.max(-5, 10);
let d = Math.max(-5, -10);
let e = Math.max(1.5, 2.5);
```

Try it Yourself »

Description

SHARE

The `Math.max()` method returns the number with the highest value.

See Also:

The Math.min() Method

REMOVE ADS

Syntax

```
Math.max(n1, n2,...)
```

Parameters

Parameter	Description
n1, n2,...	Optional. One or more numbers to compare.

Return Value

Type	Description
Number	The highest number of the arguments. `-Infinity` if no arguments are given. `NaN` if one of the arguments is not a number.

See Also:

Full JavaScript Math Tutorial

Full JavaScript Math Reference

Full JavaScript Numbers Tutorial

Full JavaScript Number Reference

Browser Support

`Math.max()` is an ECMAScript1 (JavaScript 1997) feature.

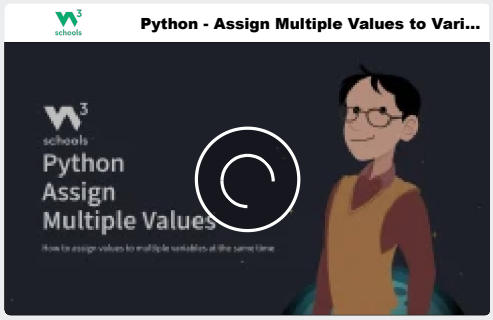
It is supported in all browsers:

Chrome	Edge	Firefox	Safari	Opera

‹ Previous | JavaScript Math Object | Next ›



REMOVE ADS



FREE ACCOUNT

Track your progress

XP earned 0

Day streak 0

Spaces 0

NEW

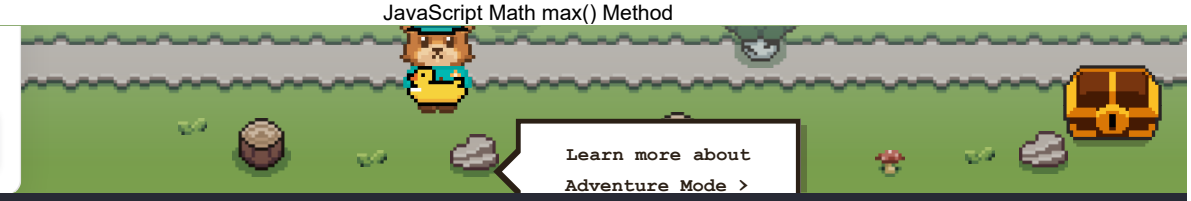
Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store | GET IT ON Google Play | TRY NOW IN Browser

League —

Sign in to track progress



PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Math.min()

‹ Previous JavaScript Math Object Next ›

Example

```
let x = Math.min(5, 10);
```

Try it Yourself »

More Examples

```
let a = Math.min(5, 10);
let b = Math.min(0, 150, 30, 20, 38);
let c = Math.min(-5, 10);
let d = Math.min(-5, -10);
let e = Math.min(1.5, 2.5);
```

Try it Yourself »

Description

SHARE

The `Math.min()` method returns the number with the lowest value.

See Also:

The Math.max() Method

REMOVE ADS

Syntax

```
Math.min(n1, n2,...)
```

Parameters

Parameter	Description
n1, n2,...	Optional. One or more numbers to compare.

Return Value

Type	Description
Number	The lowest number of the arguments. `Infinity` if no arguments are given. `NaN` if one of the arguments is not a number.

See Also:

Full JavaScript Math Tutorial

Full JavaScript Math Reference

Full JavaScript Numbers Tutorial

Full JavaScript Number Reference

Browser Support

`Math.min()` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

Chrome	Edge	Firefox	Safari	Opera

‹ Previous | JavaScript Math Object | Next ›

Learn it.
Get certified.
Get hired.

$95 per course
one-time payment

Exam and certificate included.

Get started

REMOVE ADS



FREE ACCOUNT
Track your progress

XP earned 0
Day streak 0
Spaces 0

NEW
Learn in Adventure Mode
Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store | GET IT ON Google Play | TRY NOW IN Browser



PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Math.PI

‹ Previous JavaScript Math Object Next ›

Example

```
let x = Math.PI;
```

Try it Yourself »

Description

SHARE

Math.PI returns PI (the ratio of a circle's area to the square of its radius, approximately 3.14)

Browser Support

`Math.PI` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

Chrome	Edge	Firefox	Safari	Opera

Syntax

```
Math.PI
```

Return Value

A number representing PI.

See Also:

Full JavaScript Math Tutorial

Full JavaScript Math Reference

Full JavaScript Numbers Tutorial

Full JavaScript Number Reference

‹ Previous | JavaScript Math Object | Next ›

Build skills.
Get hired.
$95 per paid course
one-time payment
Exam and certificate included.
Get started

REMOVE ADS



FREE ACCOUNT
Track your progress

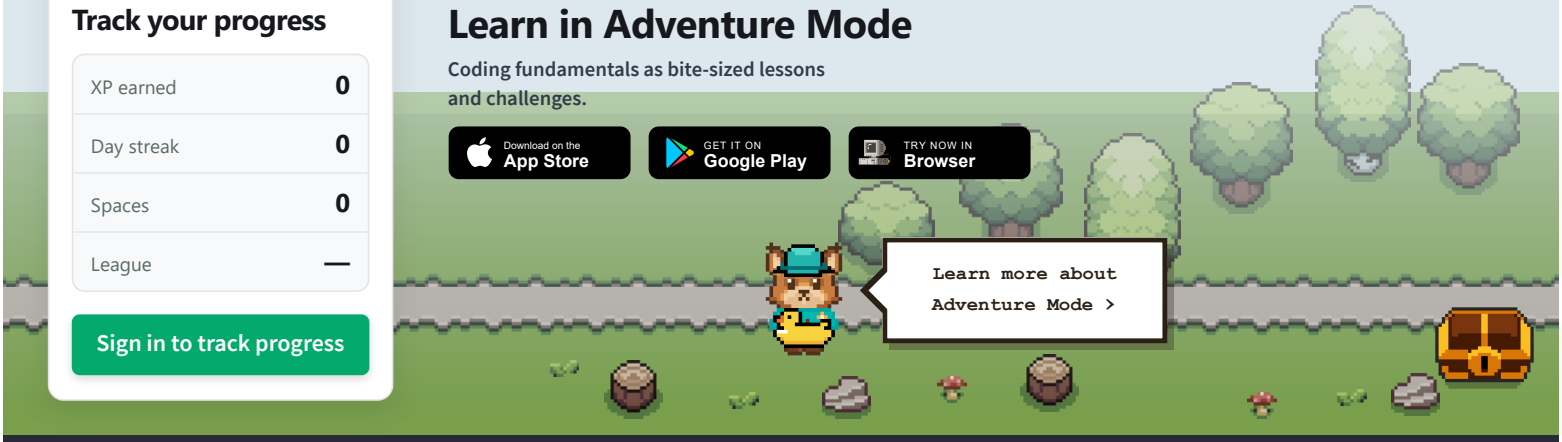
XP earned 0
Day streak 0
Spaces 0
League —

Sign in to track progress

NEW
Learn in Adventure Mode
Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store
GET IT ON Google Play
TRY NOW IN Browser

Learn more about Adventure Mode >



schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial

PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted

our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Math.pow()

‹ Previous JavaScript Math Object Next ›

Example

```
let x = Math.pow(4, 3);
```

Try it Yourself »

Description

SHARE

The Math.pow() method returns the value of x to the power of y (x^y).

More Examples

```
let a = Math.pow(0, 1);
let b = Math.pow(1, 1);
let c = Math.pow(1, 10);
let d = Math.pow(3, 3);
let e = Math.pow(-3, 3);
let f = Math.pow(2, 4);
```

Try it Yourself »

Browser Support

`Math.pow()` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

				
Chrome	Edge	Firefox	Safari	Opera

REMOVE ADS

Syntax

```
Math.pow(x, y)
```

Parameters

Parameter	Description
x	Required. The base
y	Required. The exponent

Return Value

A number representing the value of x to the power of y (x^y).

See Also:

Full JavaScript Math Tutorial

Full JavaScript Math Reference

Full JavaScript Numbers Tutorial

Full JavaScript Number Reference

‹ Previous JavaScript Math Object Next ›

Get Certified in JavaScript

Boost your career with an official certification

$95

Get certified

REMOVE ADS



FREE ACCOUNT

Track your progress

XP earned 0

Day streak 0

Spaces 0

League —

Sign in to track progress

NEW

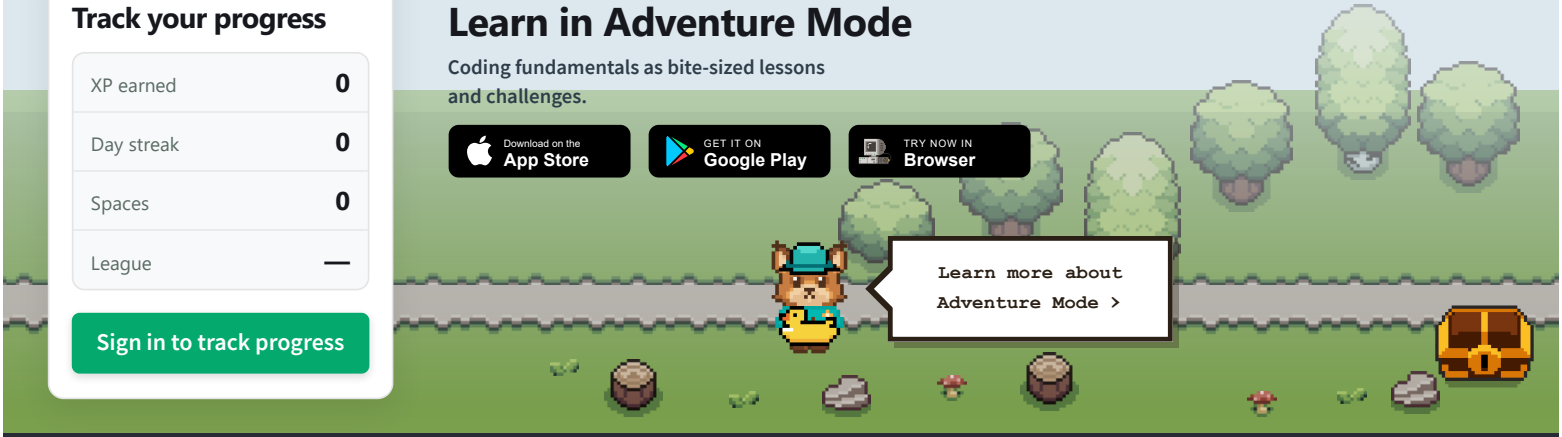
Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store

GET IT ON Google Play

TRY NOW IN Browser



PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples

Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Math.random()

‹ Previous JavaScript Math Object Next ›

Examples

```
let x = Math.random();
```

Try it Yourself »

Return a random number between 0 (inclusive) and 10 (exclusive):

```
let x = Math.random() * 10;
```

Try it Yourself »

Return a random number between 0 (inclusive) and 100 (exclusive):

```
let x = Math.random() * 100;
```

Try it Yourself »

A random whole number between 1 and 10 (inclusive):

```
let x = Math.floor((Math.random() * 10) + 1);
```

Try it Yourself »

A random whole number between 1 and 100 (inclusive):

```
let x = Math.floor((Math.random() * 100) + 1);
```

Try it Yourself »

Description

SHARE

The `Math.random()` method returns a random floating point number between 0 (inclusive) and 1 (exclusive).

Note

Math.random() does not return a cryptographically secure number.

If you need a cryptographically secure number, use this Crypto API method:
crypto.getRandomValues()

REMOVE ADS

Syntax

```
Math.random()
```

Parameters

NONE

Return Value

Type	Description
Number	A random number from 0 (inclusive) up to but not including 1 (exclusive).

See Also:

Full JavaScript Math Tutorial

Full JavaScript Math Reference

Full JavaScript Numbers Tutorial

Full JavaScript Number Reference

Browser Support

`Math.random()` is an ECMAScript1 (JavaScript 1997) feature.

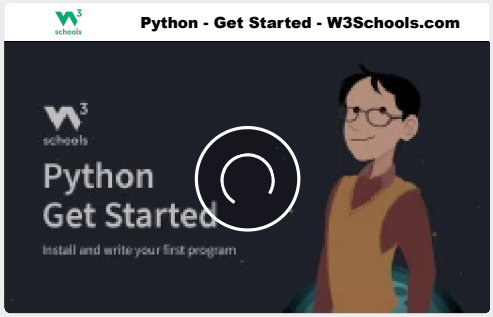
It is supported in all browsers:

Chrome	Edge	Firefox	Safari	Opera

‹ Previous | JavaScript Math Object | Next ›



REMOVE ADS



Track your progress

XP earned 0

Day streak 0

Spaces 0

League —

Sign in to track progress

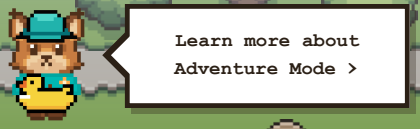
Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store

GET IT ON Google Play

TRY NOW IN Browser



W3schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference

Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate

FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Math.round()

‹ Previous JavaScript Math Object Next ›

Examples

```
let x = Math.round(2.5);
```

Try it Yourself »

```
let a = Math.round(2.60);
let b = Math.round(2.50);
let c = Math.round(2.49);
let d = Math.round(-2.60);
let e = Math.round(-2.50);
let f = Math.round(-2.49);
```

Try it Yourself »

Description

SHARE

The `Math.round()` method rounds a number to the nearest integer.

2.49 will be rounded down (2), and 2.5 will be rounded up (3).

JavaScript Rounding Functions

The Math.abs() Method
The Math.ceil() Method
The Math.floor() Method
The Math.round() Method
The Math.fround() Method
The Math.f16round() Method
The Math.trunc() Method

REMOVE ADS

Syntax

```
Math.round(x)
```

Parameters

Parameter	Description
x	Required. A number.

Return Value

Type	Description
Number	The number rounded to the nearest integer.

Browser Support

`Math.round()` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

				
Chrome	Edge	Firefox	Safari	Opera

‹ Previous | JavaScript Math Object | Next ›

Build skills.
Get hired.

$95 per paid course
one-time payment

Exam and certificate included.

Get started

REMOVE ADS

FREE ACCOUNT
Track your progress

XP earned 0
Day streak 0
Spaces 0
League —

Sign in to track progress

NEW
Learn in Adventure Mode
Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store
GET IT ON Google Play
TRY NOW IN Browser



schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate

Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate

FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Math.sign()

‹ Previous JavaScript Math Object Next ›

Example 1

```
let x = Math.sign(3);
```

Try it Yourself »

Description

SHARE

The `Math.sign()` method returns whether a number is negative, positive or zero.

If the number is positive, this method returns 1.
If the number is negative, it returns -1.
If the number is zero, it returns 0.

Example 2

```
let x = Math.sign(-3);
```

Try it Yourself »

Example 3

```
let x = Math.sign(0);
```

Try it Yourself »

Browser Support

`Math.sign()` is an ECMAScript6 (ES6 2015) feature.

JavaScript 2015 is supported in all browsers since **June 2017**:

Chrome	Edge	Firefox	Safari	Opera
51	15	54	10	38
May 2016	Apr 2017	Jun 2017	Sep 2016	Jun 2016

Syntax

```
Math.sign(x)
```

Parameters

Parameter	Description
x	Required. A number.

Return Value

A Number, representing the sign of the specified number:

- If the number is positive, it returns 1
- If the number is negative, it returns -1
- If the number is positive zero, it returns 0
- If the number is negative zero, it returns -0
- If the number is not a number, it returns NaN

See Also:

Full JavaScript Math Tutorial

Full JavaScript Math Reference

Full JavaScript Numbers Tutorial

Full JavaScript Number Reference

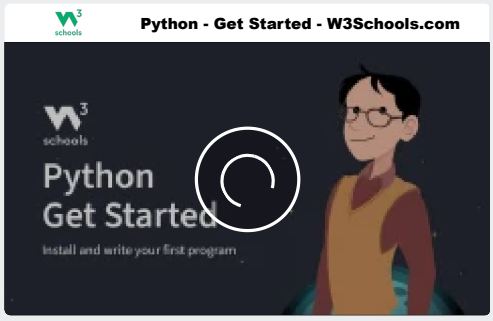
‹ Previous

JavaScript Math Object

Next ›



REMOVE ADS



FREE ACCOUNT

Track your progress

XP earned 0

NEW

Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Day streak 0
Spaces 0
League —
Sign in to track progress

Download on the App Store
GET IT ON Google Play
TRY NOW IN Browser

Learn more about Adventure Mode >



PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference

AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Math.sin()

‹ Previous JavaScript Math Object Next ›

Examples

```
let x = Math.sin(3.14);
```

Try it Yourself »

```
Math.sin(0);
Math.sin(Math.PI/6);
Math.sin(Math.PI/2);
Math.sin(Math.PI);
```

Try it Yourself »

Math.sin(x) expect x in radians.

To use degrees, convert degrees to radians first.

```
let radians = degrees * Math.PI/180;
Math.sin(radians);
```

Try it Yourself »

Description

SHARE

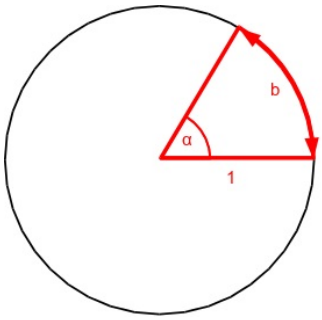
The `Math.sin()` method returns the sine of a number.

The `Math.sin()` method returns a number between -1 and 1.

The `Math.sin()` method expects the number in radians.

What is Radians?

Radians is an angle's **α** amount of rotation **b** on a circle:



Degrees	PI	Radians
0°	0	0
1°	PI/180	0.0175
30°	PI/6	0.52
45°	PI/4	0.79
90°	PI/2	1.57
180°	PI	3.14
360°	PI*2	6.28

JavaScript Sine and Cosine Methods:

The Math.sin() Method
The Math.sinh() Method
The Math.asin() Method
The Math.asinh() Method

The Math.cos() Method
The Math.cosh() Method
The Math.acos() Method
The Math.acosh() Method

REMOVE ADS

Syntax

```
Math.sin(x)
```

Parameters

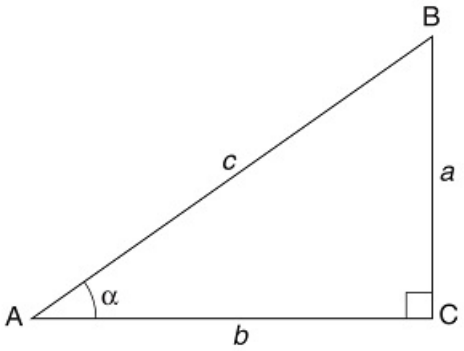
Parameter	Description
x	Required. A number representing radians.

Return Value

Type	Description
Number	-1 to 1. The sine of the number. `NaN` if the parameter is not numeric.

The Pythagoran Therorem

`Math.sin()`, `Math.cos()`, and `Math.tan()` are related to the Pythagorean theorem:



Theorem: $c^2 = a^2 + b^2$

The sine `Math.sin()` to the angle is **a / c**.

The cosine `Math.cos()` to the angle is **b / c**.

The tangent `Math.tan()` to the angle is **a / b**.

See Also:

Full JavaScript Math Tutorial

Full JavaScript Math Reference

Full JavaScript Numbers Tutorial

Full JavaScript Number Reference

Browser Support

`Math.sin()` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

Chrome	Edge	Firefox	Safari	Opera

‹ Previous JavaScript Math Object Next ›



REMOVE ADS

Python - Introduction - W3Schools.com



FREE ACCOUNT

Track your progress

XP earned 0

Day streak 0

Spaces 0

League —

Sign in to track progress

NEW

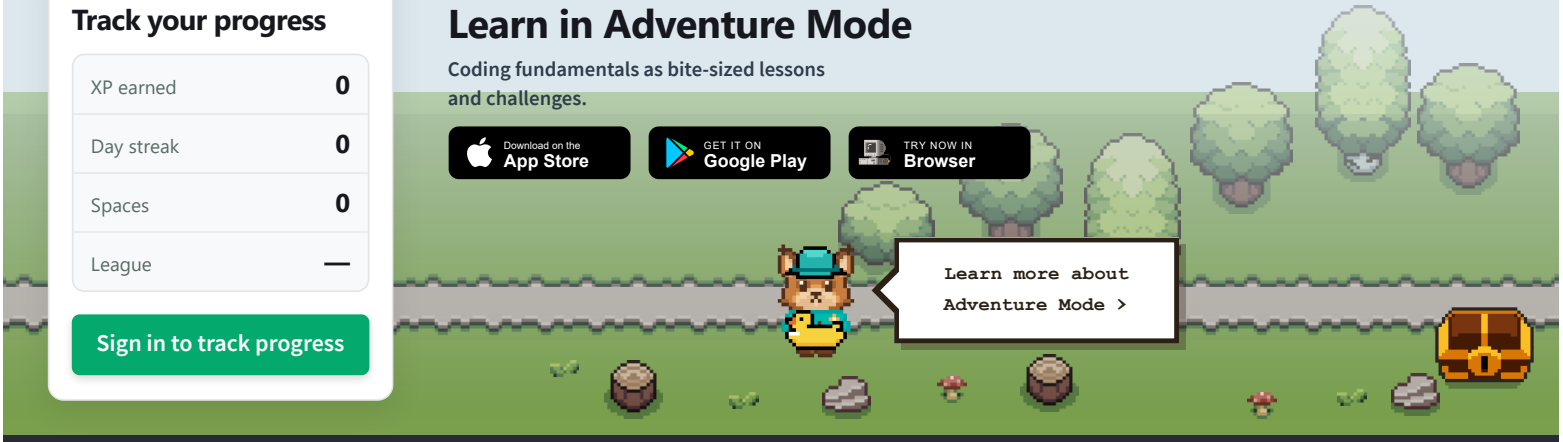
Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store

GET IT ON Google Play

TRY NOW IN Browser



schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial

PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted

our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Math.sinh()

‹ Previous JavaScript Math Object Next ›

Example

Return the hyperbolic sine of a number:

```
Math.sinh(3);
```

Try it Yourself »

Description

SHARE

The `Math.sinh()` method returns the hyperbolic sine of a number.

JavaScript Sine and Cosine Methods:

The Math.sin() Method
The Math.sinh() Method
The Math.asin() Method
The Math.asinh() Method

The Math.cos() Method
The Math.cosh() Method

The Math.acos() Method
The Math.acosh() Method

REMOVE ADS

Syntax

```
Math.sinh(x)
```

Parameters

Parameter	Description
x	Required. A number.

Return Value

Type	Description
Number	The hyperbolic sine of the number. `NaN` if the parameter is not numeric.

See Also:

Full JavaScript Math Tutorial

Full JavaScript Math Reference

Full JavaScript Numbers Tutorial

Full JavaScript Number Reference

Browser Support

`Math.sinh` is an ECMAScript6 (ES6 2015) feature.

JavaScript 2015 is supported in all browsers since **June 2017**:

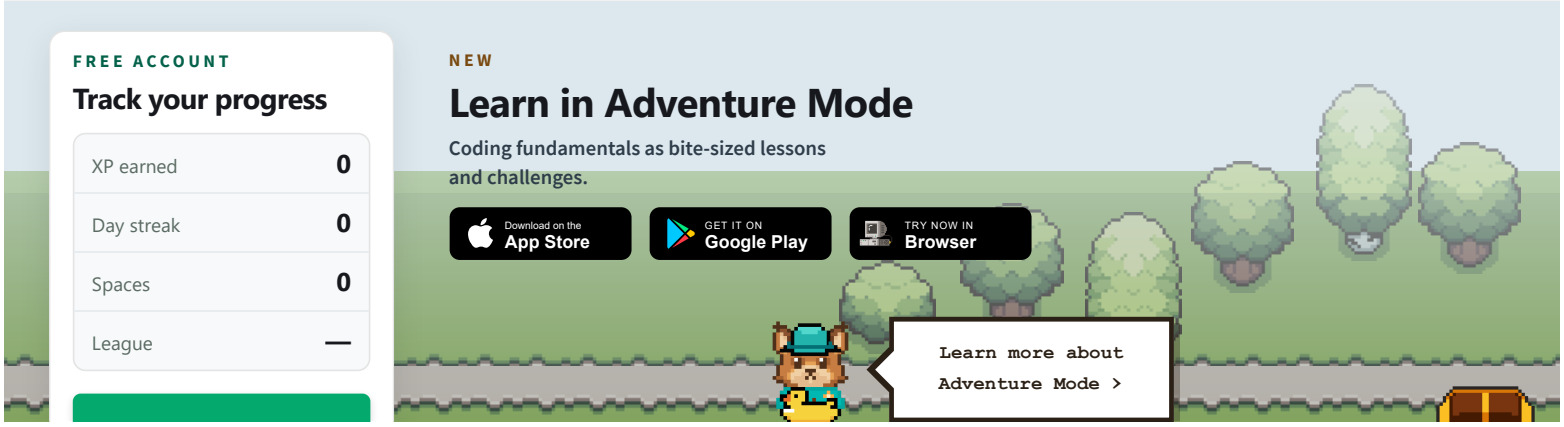
Chrome	Edge	Firefox	Safari	Opera
51	15	54	10	38
May 2016	Apr 2017	Jun 2017	Sep 2016	Jun 2016

‹ Previous | JavaScript Math Object | Next ›

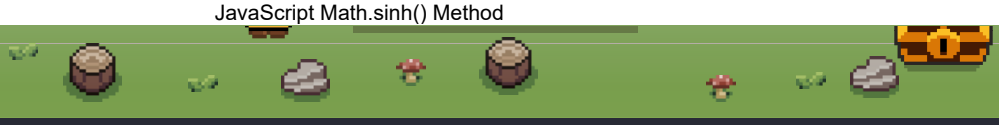


REMOVE ADS





Sign in to track progress



PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate

FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Math.sqrt()

‹ Previous | JavaScript Math Object | Next ›

Examples

```
let x = Math.sqrt(9);
```

Try it Yourself »

```
let a = Math.sqrt(0);
let b = Math.sqrt(1);
let c = Math.sqrt(9);
let d = Math.sqrt(64);
let e = Math.sqrt(-9);
```

Try it Yourself »

Description

SHARE

The `Math.sqrt()` method returns the square root of a number.

See Also:

The Math.cbrt() Method
The Math.SQRT2 Property

The Math.SQRT1_2 Property

REMOVE ADS

Syntax

```
Math.sqrt(x)
```

Parameters

Parameter	Description
x	Required. A number.

Return Value

Type	Description
Number	The square root of the number. `NaN` if the number negative.

See Also:

Full JavaScript Math Tutorial

Full JavaScript Math Reference

Full JavaScript Numbers Tutorial

Full JavaScript Number Reference

Browser Support

`Math.` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

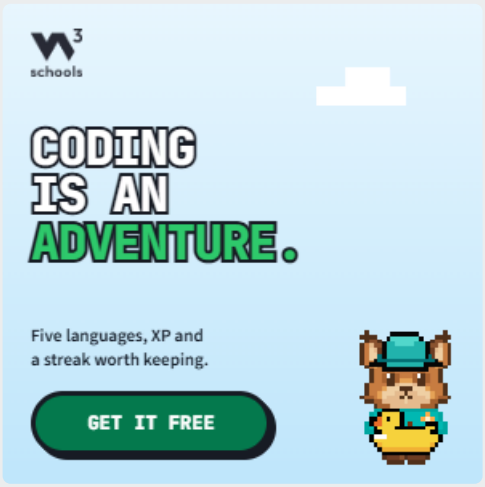






Chrome	Edge	Firefox	Safari	Opera

‹ Previous | JavaScript Math Object | Next ›



REMOVE ADS



FREE ACCOUNT

Track your progress

XP earned	0
Day streak	0
Spaces	0
League	—

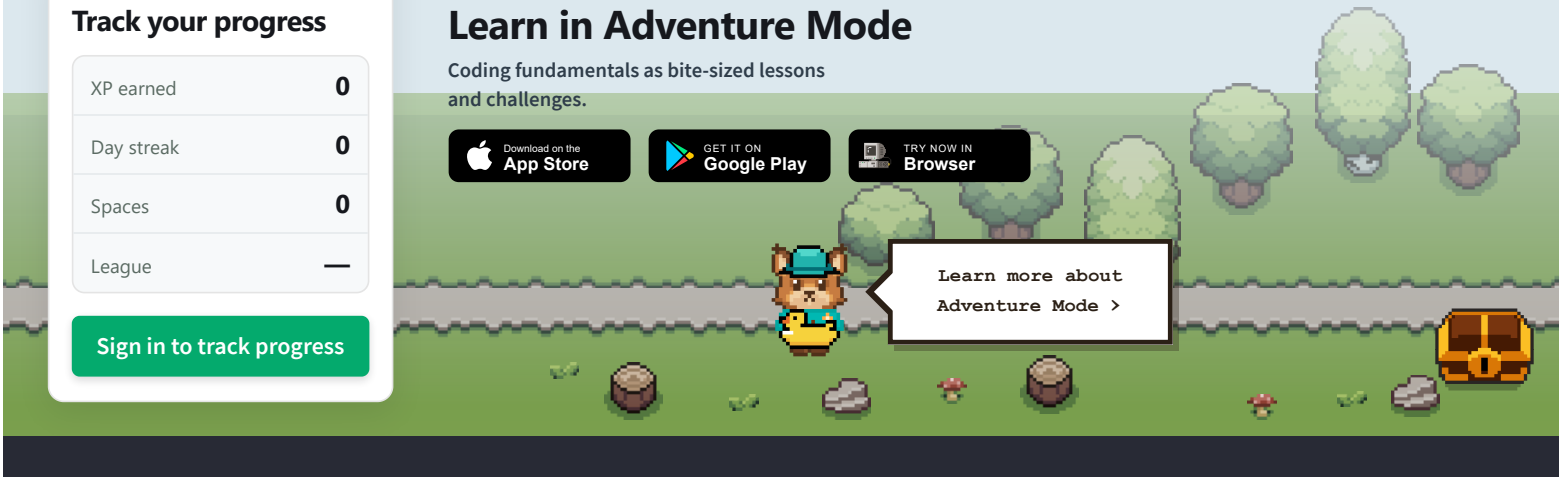
Sign in to track progress

NEW

Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store | GET IT ON Google Play | TRY NOW IN Browser



schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples

SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate

FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Map values()

‹ Previous JavaScript Map Reference Next ›

Map.values()

SHARE

Example

```
// Create a Map
const fruits = new Map([
  ["apples", 500],
  ["bananas", 300],
  ["oranges", 200]
]);

// List all values
let text = "";
for (const x of fruits.values()) {
  text += x;
}
```

Try it Yourself »

More Examples Below !

Description

The `values()` method returns an iterator object with the values in a map.

The `values()` method does not change the original map.

Syntax

```
map.values()
```

Parameters

NONE

Return Value

Type	Description
Iterator	An iterable object with the values of the map.

REMOVE ADS

Related Pages:

JavaScript Maps

JavaScript Iterables

Full JavaScript Map Reference

More Examples

Use the `values()` method to sum the values in a Map:

Example

```
// Create a Map
const fruits = new Map([
  ["apples", 500],
  ["bananas", 300],
  ["oranges", 200]
]);
```

```
// Sum all Values
let total = 0;
for (const x of fruits.values()) {
  total += x;
}
```

Try it Yourself »

Browser Support

`map.values()` is an ECMAScript6 (ES6 2015) feature.

JavaScript 2015 is supported in all browsers since **June 2017**:

Chrome	Edge	Firefox	Safari	Opera
51	15	54	10	38
May 2016	Apr 2017	Jun 2017	Sep 2016	Jun 2016

❮ Previous JavaScript Map Reference Next ❯



REMOVE ADS

Python - Get Started - W3Schools.com



FREE ACCOUNT

Track your progress

XP earned 0

Day streak 0

Spaces 0

League —

Sign in to track progress

NEW

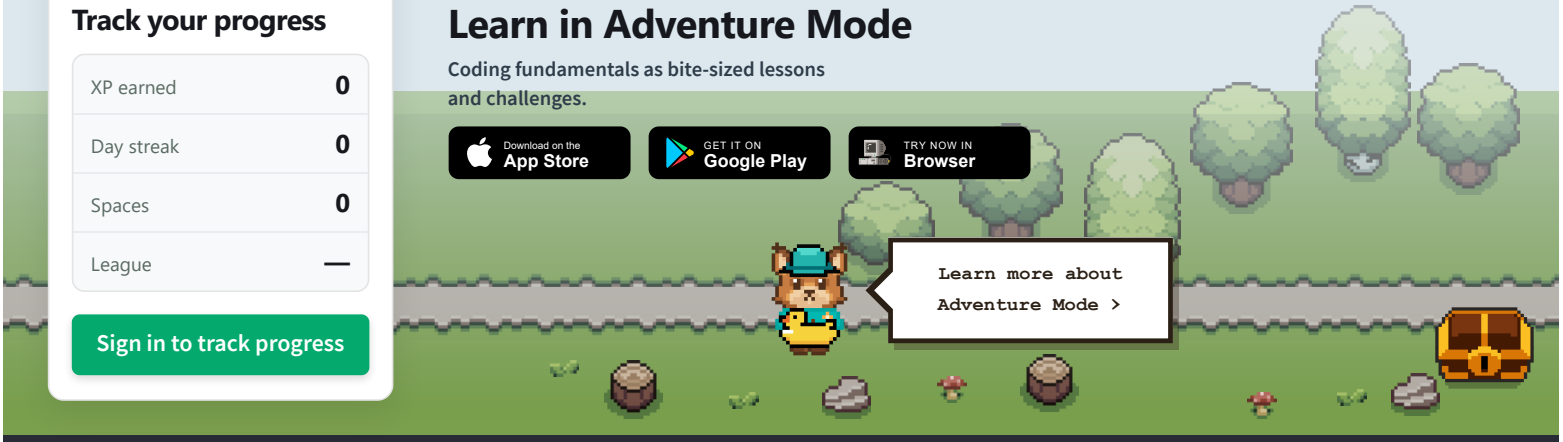
Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store

GET IT ON Google Play

TRY NOW IN Browser



schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial

PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted

our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Math.abs()

‹ Previous JavaScript Math Object Next ›

Examples

```
let x = Math.abs(-7.25);
```

Try it Yourself »

```
let a = Math.abs(7.25);
let b = Math.abs(-7.25);
let c = Math.abs(null);
let d = Math.abs("Hello");
let e = Math.abs(2-3);
```

Try it Yourself »

Description

SHARE

The `Math.abs()` method returns the absolute value of a number.

JavaScript Rounding Functions

The Math.abs() Method
The Math.ceil() Method

The Math.floor() Method
The Math.round() Method
The Math.fround() Method
The Math.f16round() Method
The Math.trunc() Method

REMOVE ADS

Syntax

```
Math.abs(x)
```

Parameters

Parameter	Description
x	Required. A number.

Return Value

Type	Description
Number	The absolute value of the number. `NaN` if the value is not a number. 0 if the value is `null`.

See Also:

Full JavaScript Math Tutorial

Full JavaScript Math Reference

Full JavaScript Numbers Tutorial

Full JavaScript Number Reference

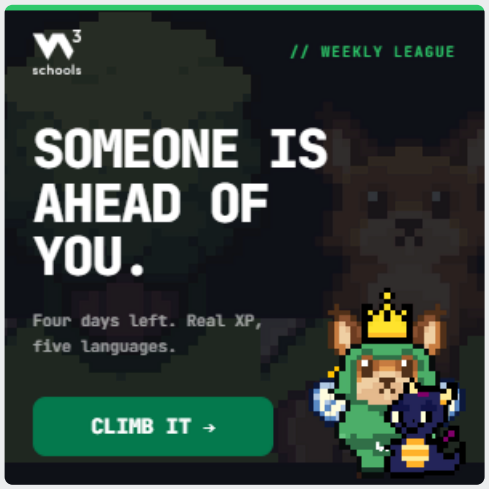
Browser Support

`Math.abs()` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

Chrome	Edge	Firefox	Safari	Opera

‹ Previous | JavaScript Math Object | Next ›



REMOVE ADS



FREE ACCOUNT

Track your progress

XP earned 0

Day streak 0

Spaces 0

NEW

Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store | GET IT ON Google Play | TRY NOW IN Browser



PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Math.SQRT1_2

‹ Previous JavaScript Math Object Next ›

Example

```
let x = Math.SQRT1_2;
```

Try it Yourself »

Description

SHARE

The `Math.SQRT1_2` property returns the square root of 1/2.

The `Math.SQRT1_2` property returns approximately 0.707.

See Also:

The Math.sqrt() Method
The Math.cbrt() Method
The Math.SQRT2 Property

Syntax

```
Math.SQRT1_2
```

Return Value

Type	Description
Number	The square root of 1/2.

See Also:

Full JavaScript Math Tutorial

Full JavaScript Math Reference

Full JavaScript Numbers Tutorial

Full JavaScript Number Reference

Browser Support

`Math.SQRT1_2` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

				
Chrome	Edge	Firefox	Safari	Opera

‹ Previous JavaScript Math Object Next ›



REMOVE ADS



FREE ACCOUNT

Track your progress

XP earned	0
Day streak	0
Spaces	0
League	—

Sign in to track progress

NEW

Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store | GET IT ON Google Play | TRY NOW IN Browser



schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate

FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Math.SQRT2

‹ Previous JavaScript Math Object Next ›

Example

```
let x = Math.SQRT2;
```

Try it Yourself »

Description

SHARE

The `Math.SQRT2` property returns the square root of 2.

The `Math.SQRT2` property returns approximately 1.414.

See Also:

The Math.sqrt() Method
The Math.cbrt() Method
The Math.SQRT1_2 Property

Syntax

```
Math.SQRT2
```

Return Value

Type	Description
Number	The square root of 2.

See Also:

Full JavaScript Math Tutorial

Full JavaScript Math Reference

Full JavaScript Numbers Tutorial

Full JavaScript Number Reference

Browser Support

`Math.SQRT2` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

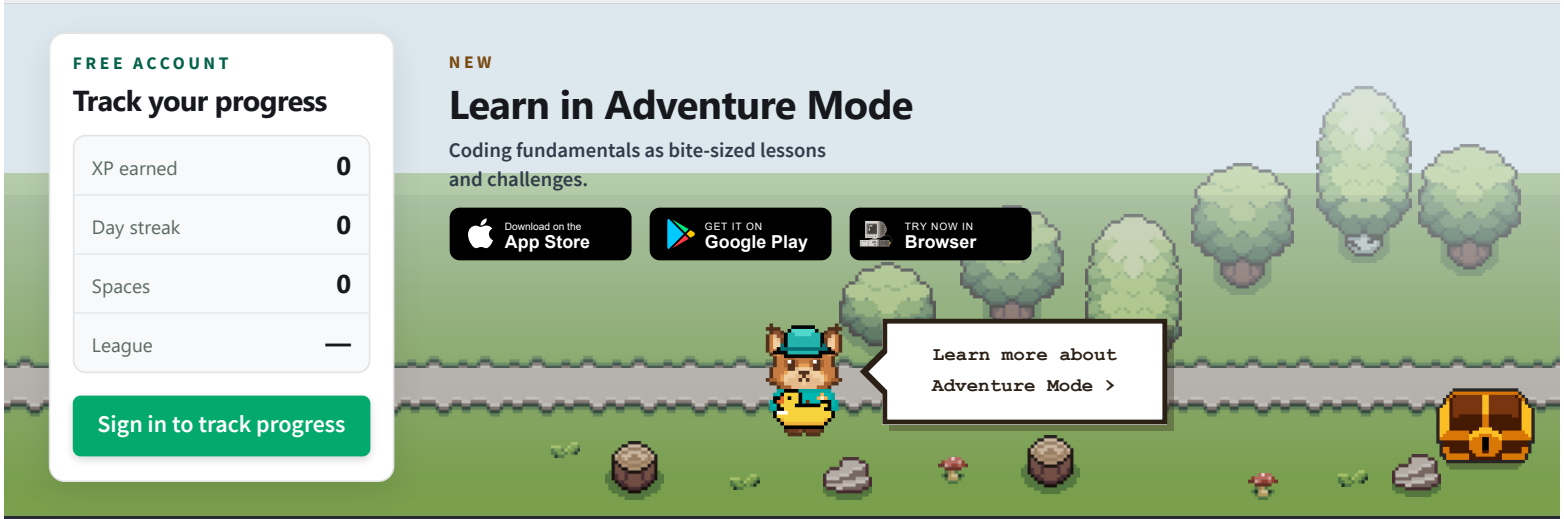
Chrome	Edge	Firefox	Safari	Opera

‹ Previous JavaScript Math Object Next ›



REMOVE ADS





schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate

FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Math.tan()

‹ Previous JavaScript Math Object Next ›

Examples

```
let x = Math.tan(1);
```

Try it Yourself »

```
Math.tan(0);
Math.tan(Math.PI);
```

Try it Yourself »

Math.tan(x) expect x in radians.

To use degrees, convert degrees to radians first.

```
let radians = degrees * Math.PI/180;
Math.tan(radians);
```

Try it Yourself »

Description

SHARE

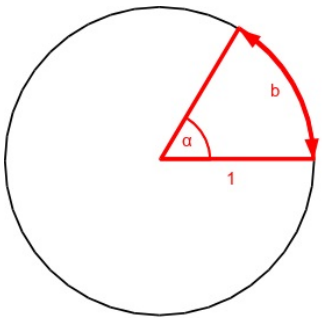
The `Math.tan()` method returns the tangent of a number.

The `Math.tan()` method expects the number in radians.

Based on sine and cosine:
`Math.tan(x) = Math.sin(x) / Math.cos(x)`.

What is Radians?

Radians is an angle's **α** amount of rotation **b** on a circle:



Degrees	PI	Radians
0°	0	0
1°	PI/180	0.0175
30°	PI/6	0.52
45°	PI/4	0.79
90°	PI/2	1.57
180°	PI	3.14
360°	PI*2	6.28

JavaScript Tangent Methods:

The Math.tan() Method
The Math.tanh() Method
The Math.atan() Method
The Math.atan2() Method
The Math.atanh() Method

Sine and Cosine Methods:

The Math.sin() Method
The Math.cos() Method

REMOVE ADS

Syntax

```
Math.tan(x)
```

Parameters

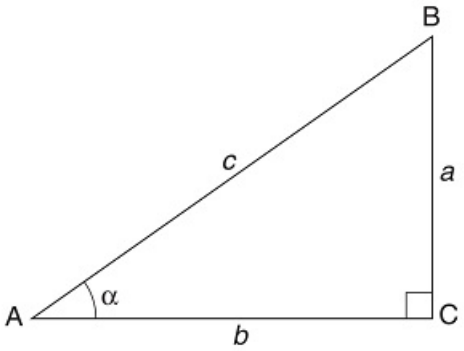
Parameter	Description
x	Required. A number representing radians.

Return Value

Type	Description
Number	The tangent of the number. `NaN` if the number is empty or not numeric.

The Pythagoran Therorem

`Math.sin()`, `Math.cos()`, and `Math.tan()` are related to the Pythagorean theorem:



Theorem: $c^2 = a^2 + b^2$

The sine `Math.sin()` to the angle is **a / c**.

The cosine `Math.cos()` to the angle is **b / c**.

The tangent `Math.tan()` to the angle is **a / b**.

See Also:

Full JavaScript Math Tutorial

Full JavaScript Math Reference

Full JavaScript Numbers Tutorial

Full JavaScript Number Reference

Browser Support

`Math.tan()` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

Chrome	Edge	Firefox	Safari	Opera

❮ Previous JavaScript Math Object Next ❯



REMOVE ADS

Python - Introduction - W3Schools.com



FREE ACCOUNT

Track your progress

XP earned 0

Day streak 0

Spaces 0

League —

Sign in to track progress

NEW

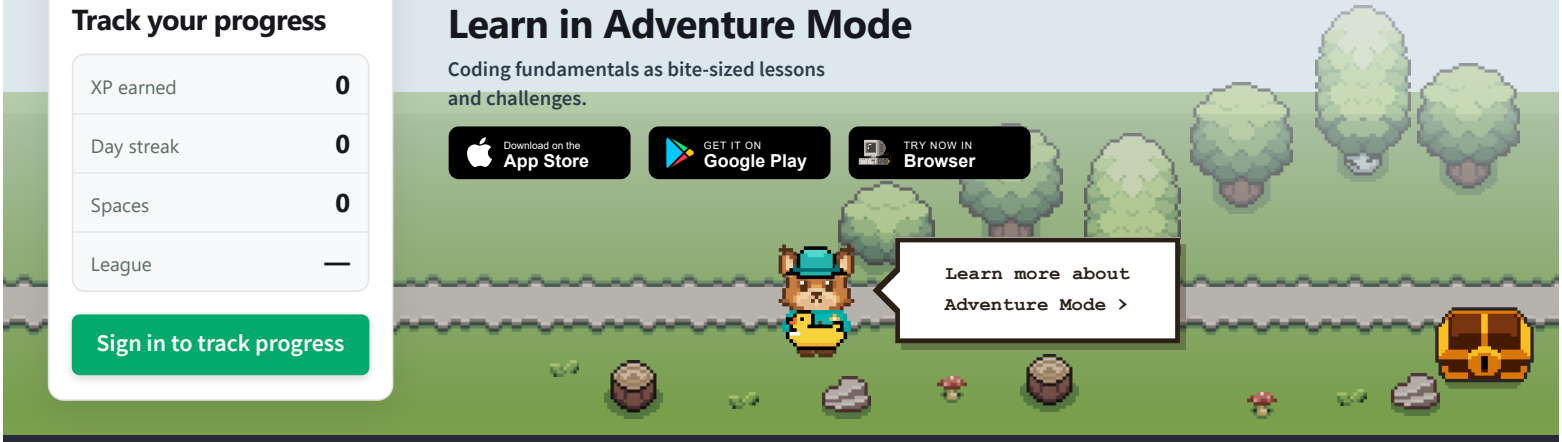
Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store

GET IT ON Google Play

TRY NOW IN Browser



schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial

PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted

our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Math.tanh()

‹ Previous JavaScript Math Object Next ›

Example

```
let x = Math.tanh(1);
```

Try it Yourself »

Description

SHARE

The `Math.tanh()` method returns the hyperbolic tangent of a number.

Based on hyperbolic sine and cosine:
`Math.tanh(x) = Math.sinh(x) / Math.cosh(x)`.

JavaScript Tangent Methods:

The Math.tan() Method
The Math.tanh() Method
The Math.atan() Method
The Math.atan2() Method
The Math.atanh() Method

Sine and Cosine Methods:

The Math.sin() Method
The Math.cos() Method

Syntax

```
Math.tanh(x)
```

Parameters

Parameter	Description
x	Required. A number.

Return Value

Type	Description
Number	The hyperbolic tangent of the number.

REMOVE ADS

See Also:

Full JavaScript Math Tutorial

Full JavaScript Math Reference

Full JavaScript Numbers Tutorial

Full JavaScript Number Reference

Browser Support

`Math.` is an ECMAScript6 (ES6 2015) feature.

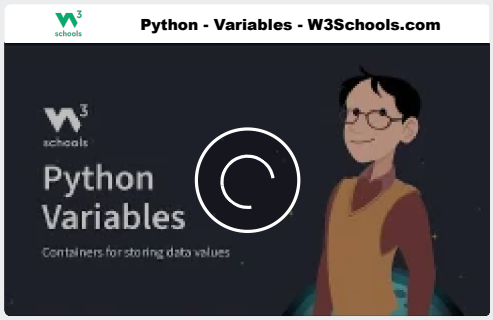
JavaScript 2015 is supported in all browsers since **June 2017**:

Chrome 51	Edge 15	Firefox 54	Safari 10	Opera 38
May 2016	Apr 2017	Jun 2017	Sep 2016	Jun 2016

‹ Previous | JavaScript Math Object | Next ›



REMOVE ADS



FREE ACCOUNT
Track your progress
XP earned 0
Day streak 0
Spaces 0
League —

NEW
Learn in Adventure Mode
Coding fundamentals as bite-sized lessons and challenges.
Download on the App Store | GET IT ON Google Play | TRY NOW IN Browser



Sign in to track progress



PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples

How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate

FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Math.trunc()

‹ Previous | JavaScript Math Object | Next ›

Example

```
let x = Math.trunc(8.76);
```

Try it Yourself »

Description

SHARE

The `Math.trunc()` method returns the integer part of a number.

The `Math.trunc()` method removes the decimals (does NOT round the number).

JavaScript Rounding Functions

The Math.abs() Method
The Math.ceil() Method
The Math.floor() Method
The Math.round() Method
The Math.fround() Method
The Math.f16round() Method
The Math.trunc() Method

Syntax

```
Math.trunc(x)
```

Parameters

Parameter	Description
x	Required. A number.

Return Value

Type	Description
Number	The integer part of the number.

See Also:

Full JavaScript Math Tutorial

Full JavaScript Math Reference

Full JavaScript Numbers Tutorial

Full JavaScript Number Reference

Browser Support

`Math.trunc()` is an ECMAScript6 (ES6 2015) feature.

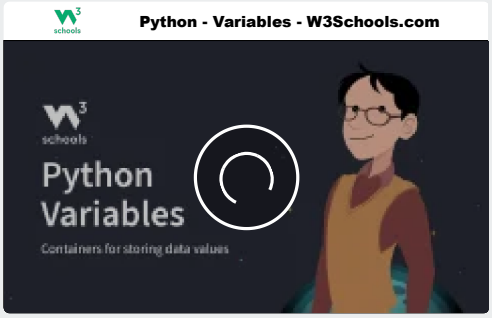
JavaScript 2015 is supported in all browsers since **June 2017**:

				
Chrome 51	Edge 15	Firefox 54	Safari 10	Opera 38
May 2016	Apr 2017	Jun 2017	Sep 2016	Jun 2016

‹ Previous JavaScript Math Object Next ›



REMOVE ADS





schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples

Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

JavaScript Number Reference

‹ Previous

Next ›

New to JavaScript Numbers?

SHARE

Visit Our Full JavaScript Number Tutorial

Complete Number Reference

Revised July 2025

Name	Description
constructor	Returns the function that created JavaScript's Number prototype
EPSILON	Returns the difference between 1 and the smallest number greater than 1
isFinite()	Checks whether a value is a finite number
isInteger()	Checks whether a value is an integer
isNaN()	Checks whether a value is Number.NaN
isSafeInteger()	Checks whether a value is a safe integer
MAX_SAFE_INTEGER	Returns the maximum safe integer in JavaScript.
MIN_SAFE_INTEGER	Returns the minimum safe integer in JavaScript
MAX_VALUE	Returns the largest number possible in JavaScript

MIN_VALUE	Returns the smallest number possible in JavaScript
NaN	Represents a "Not-a-Number" value
NEGATIVE_INFINITY	Represents negative infinity (returned on overflow)
POSITIVE_INFINITY	Represents infinity (returned on overflow)
parseFloat()	Parses a string an returns a number
parseInt()	Parses a string an returns a whole number
prototype	Allows you to add properties and methods to an object
toExponential(x)	Converts a number into an exponential notation
toFixed(x)	Formats a number with x numbers of digits after the decimal point
toLocaleString()	Converts a number into a string, based on the locale settings
toPrecision(x)	Formats a number to x length
toString()	Converts a number to a string
valueOf()	Returns the primitive value of a number

NOTE

All number methods return a new value.

They do not change the original number.

See Also:

Full JavaScript Number Tutorial

Full JavaScript Math Tutorial

Full JavaScript Math Reference

REMOVE ADS

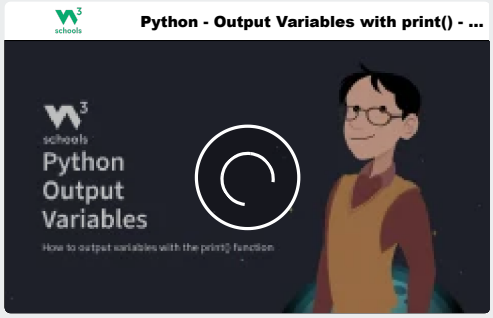
‹ Previous

Sign in to track progress

Next ›



REMOVE ADS



FREE ACCOUNT

Track your progress

XP earned	0
Day streak	0
Spaces	0
League	—

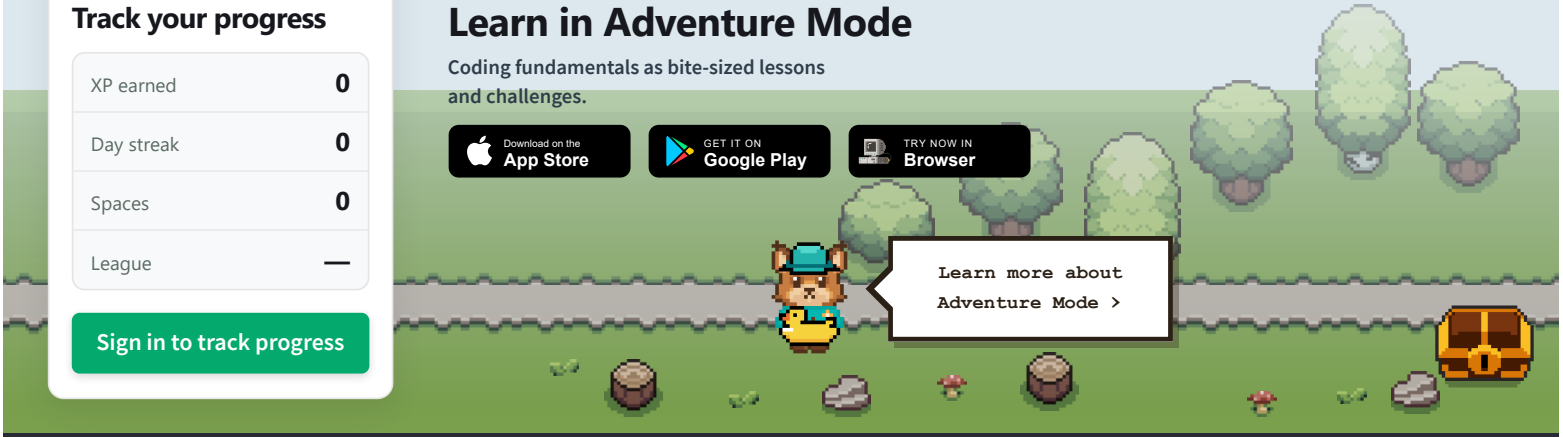
Sign in to track progress

NEW

Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store | GET IT ON Google Play | TRY NOW IN Browser



schools

PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples

Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.

Complete JavaScript References

‹ Home

Next ›

Revised June 2026

JavaScript Primitives

SHARE

Primitives | String | Number | Boolean | BigInt | Symbol | undefined | null

Object Methods and Properties

Array | Boolean | Classes | Date | Error | Function | Global | Iterator | JSON | Map | Math | Number | Object | Operators | Precedence | Promise | Proxy | Reflect | RegExp | Set | Statements | String | Temporal | Typed Array

Window Objects with Examples

Window History Location Navigator Screen

HTML DOM Objects with Examples

Document Element Attributes Events Event Objects Collection Style

REMOVE ADS

Web APIs with Examples

Canvas Console Fetch Fullscreen Geolocation History MediaQuery

Storage Validation

HTML Elements with Examples

a abbr address area article aside audio b base bdo blockquote body br button canvas caption

cite code col colgroup datalist dd del details dfn dialog div dl dt em embed fieldset

figcaption figure footer form head header h1 - h6 hr html i iframe img ins input button

input checkbox input color input date input datetime input datetime-local input email input file input hidden

input image input month input number input password input radio input range input reset input search input submit

input text input time input url input week kbd label legend li link map mark menu menuitem meta

meter nav object ol optgroup option output p param pre progress q s samp script section

select small source span strong style sub summary sup table tbody td tfoot th thead tr

textarea time title track u ul var video

Other

CSSStyleDeclaration JS Conversion

JavaScript Tutorial

Visit Our JavaScript Tutorial »

‹ Home | Sign in to track progress | Next ›

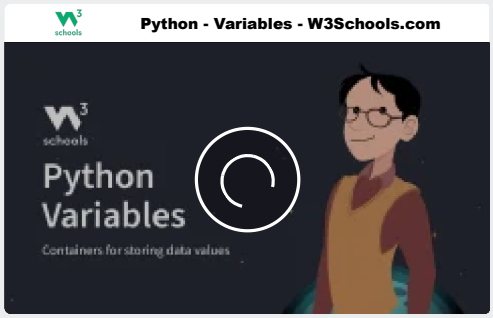
Learn it.
Get certified.
Get hired.

$95 per course
one-time payment

Exam and certificate included.

Get started

REMOVE ADS



FREE ACCOUNT
Track your progress

XP earned 0
Day streak 0
Spaces 0
League —

NEW
Learn in Adventure Mode
Coding fundamentals as bite-sized lessons and challenges.

Download on the App Store | GET IT ON Google Play | TRY NOW IN Browser

Learn more about



PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

PRACTICE

CONTACT US

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
AngularJS Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples
JavaScript Examples
How To Examples
SQL Examples
Python Examples
W3.CSS Examples
Bootstrap Examples
PHP Examples
Java Examples
XML Examples
jQuery Examples

Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate



FORUM ABOUT CAMPUS

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

Copyright 1999-2026 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.